

W4_Linear_Models_Assignment

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1 Week 4 — Statistical Machine Learning: Linear Models

1.0.1 AIF 2026 · Phase 2 · Facilitator: Raj Kumar Biswokarma

1.1 Session Roadmap

Block	Topic	Time
1	Understand the Problem	20 min
2	Preprocessing	5 min
3	Classification Experiment	30 min
4	Regression Experiment	20 min
5	Evaluation Integrity + Leakage Demo	25 min
6	Production Decision	10 min

1.2 Business Context

You are a Data Scientist at a telecom company. The business is losing customers to churn. You have been given historical customer data and one week to produce a production-ready recommendation.

There is no prescribed model. You decide — and you must defend that decision with evidence.

Dataset: Telco Customer Churn — <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Single CSV · 7043 rows · 21 columns · Target: **Churn** (Yes/No)

1.3 Important Note

This is a guided notebook. You are expected to: - Think critically before writing any code - Make decisions and justify them in writing - Explore different approaches and compare results - Fill in all written response sections

```
[1]: import numpy as np
import pandas as pd
```

```

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.dummy import DummyClassifier
from sklearn.model_selection import train_test_split, cross_val_score,
    ↳StratifiedKFold, learning_curve
from sklearn.metrics import (f1_score, accuracy_score, recall_score,
    ↳precision_score, roc_auc_score,
    ↳average_precision_score, log_loss, roc_curve, precision_recall_curve, auc,
    ↳mean_absolute_error, mean_squared_error,
    ↳r2_score, classification_report, confusion_matrix, ConfusionMatrixDisplay)
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression, SGDClassifier,
    ↳RidgeClassifier, LinearRegression, Ridge, Lasso, ElasticNet

import time

```

```

[2]: # Download Dataset

import os
import opendatasets as od

dataset_url = "https://www.kaggle.com/datasets/blastchar/telco-customer-churn"
dataset_dir = "telco-customer-churn"

# Download only if the folder does not already exist
if not os.path.exists(dataset_dir):
    od.download(dataset_url)
    print("Dataset downloaded successfully.")
else:
    print("Dataset already exists. Skipping download.")

```

Dataset already exists. Skipping download.

2 Block 1 — Understand the Problem First

2.1 What Is an ML Problem?

Before touching any model, you need to formulate the problem formally. Every ML problem has the same components:

Symbol	Name	In our problem
X	Feature space	All input columns: tenure, charges, contract type...
y	Target variable	Churn: Yes / No
H	Hypothesis class	The family of functions we search over: linear models
L	Loss function	How we measure error: cross-entropy for classification
E	Evaluation metric	How the business measures success: F1, PR-AUC

Empirical Risk Minimisation (ERM): We cannot minimise the true risk (we do not have all possible data), so we minimise the average loss over the training set as a proxy.

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n L(y_i, f_{\theta}(x_i))$$

2.2 Probability Distributions in ML

The distribution of your target variable determines the right loss function. This is not arbitrary.

Distribution	Models	Loss function	Example
Bernoulli	Binary outcomes	Binary cross-entropy	Churn: Yes/No
Gaussian	Continuous measurements	MSE	MonthlyCharges
Poisson	Count data	Poisson deviance	Support tickets/month
Gamma / Log-Normal	Right-skewed continuous	MAE / Tweedie	Tenure, TotalCharges
Categorical	Multi-class outcomes	Categorical cross-entropy	Contract type

Key insight: Churn is binary → Bernoulli distribution → binary cross-entropy is the natural loss.

2.3 Sources of Uncertainty

Training data is a finite, imperfect sample of the real world. Before modelling, identify where the data might fail you:

- **Sampling noise** — 7,043 customers is a sample. The true population may behave differently.
- **Label noise** — Was churn recorded correctly? Cancelled but re-subscribed? Edge cases.
- **Missing data** — TotalCharges has whitespace nulls. Why? Random or systematic?
- **Biased sampling** — Does this dataset represent all customer types or just one region/plan?

- **Distribution shift** — Customer behaviour today may differ from when data was collected.
- **Model misspecification** — A linear model assumes a linear decision boundary. Is that true?

2.4 1.1 Basic Inspection

2.4.1 Task:

- Load the dataset
- View the first few rows
- Check data types and shape
- Look at summary statistics

2.4.2 Questions:

- What does each row represent?
- Are the data types appropriate for each column?
- Is TotalCharges the dtype you expect?

2.4.3 Hint:

- Use `.head()`, `.info()`, `.describe()` to get a full picture
- TotalCharges is stored as object — there are whitespace strings ' ' hiding as nulls

```
[3]: # Read Data
df = pd.read_csv('telco-customer-churn/WA_Fn-UseC_-Telco-Customer-Churn.csv')
df.head()
```

```
[3]:  customerID  gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0  7590-VHVEG  Female                0      Yes           No         1           No
1  5575-GNVDE   Male                0      No            No        34           Yes
2  3668-QPYBK   Male                0      No            No         2           Yes
3  7795-CFOCW   Male                0      No            No        45           No
4  9237-HQITU   Female              0      No            No         2           Yes
```

```
    MultipleLines  InternetService  OnlineSecurity  ...  DeviceProtection  \
0  No phone service              DSL              No  ...              No
1                No              DSL              Yes  ...              Yes
2                No              DSL              Yes  ...              No
3  No phone service              DSL              Yes  ...              Yes
4                No  Fiber optic              No  ...              No
```

```
    TechSupport  StreamingTV  StreamingMovies  ...  Contract  PaperlessBilling  \
0            No            No                No  Month-to-month              Yes
1            No            No                No      One year              No
2            No            No                No  Month-to-month              Yes
3            Yes            No                No      One year              No
4            No            No                No  Month-to-month              Yes
```

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

[5 rows x 21 columns]

```
[4]: # your code here
```

```
df.describe()
```

```
[4]:
```

	SeniorCitizen	tenure	MonthlyCharges
count	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692
std	0.368612	24.559481	30.090047
min	0.000000	0.000000	18.250000
25%	0.000000	9.000000	35.500000
50%	0.000000	29.000000	70.350000
75%	0.000000	55.000000	89.850000
max	1.000000	72.000000	118.750000

```
[5]: # your code here
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   str
1   gender                 7043 non-null   str
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   str
4   Dependents             7043 non-null   str
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   str
7   MultipleLines           7043 non-null   str
8   InternetService         7043 non-null   str
9   OnlineSecurity          7043 non-null   str
10  OnlineBackup            7043 non-null   str
11  DeviceProtection        7043 non-null   str
12  TechSupport             7043 non-null   str
13  StreamingTV             7043 non-null   str
14  StreamingMovies         7043 non-null   str
15  Contract                7043 non-null   str
```

```

16 PaperlessBilling 7043 non-null str
17 PaymentMethod 7043 non-null str
18 MonthlyCharges 7043 non-null float64
19 TotalCharges 7043 non-null str
20 Churn 7043 non-null str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB

```

```
[6]: df.nunique()
```

```

[6]: customerID      7043
gender              2
SeniorCitizen      2
Partner            2
Dependents         2
tenure             73
PhoneService       2
MultipleLines      3
InternetService    3
OnlineSecurity     3
OnlineBackup       3
DeviceProtection  3
TechSupport        3
StreamingTV        3
StreamingMovies    3
Contract           3
PaperlessBilling   2
PaymentMethod      4
MonthlyCharges    1585
TotalCharges      6531
Churn              2
dtype: int64

```

```

[7]: cols = [
    ↪ "gender", "SeniorCitizen", "Partner", "Dependents", "PhoneService", "MultipleLines", "InternetService",
    ↪ "StreamingTV", "StreamingMovies", "Contract", "PaperlessBilling", "PaymentMethod", "Churn"]

for col in cols:
    print(f"Unique items in \"{col}\" column")
    print(df[col].unique().tolist(), end="\n\n")

```

```

Unique items in "gender" column
['Female', 'Male']

```

```

Unique items in "SeniorCitizen" column
[0, 1]

```

Unique items in "Partner" column
['Yes', 'No']

Unique items in "Dependents" column
['No', 'Yes']

Unique items in "PhoneService" column
['No', 'Yes']

Unique items in "MultipleLines" column
['No phone service', 'No', 'Yes']

Unique items in "InternetService" column
['DSL', 'Fiber optic', 'No']

Unique items in "OnlineSecurity" column
['No', 'Yes', 'No internet service']

Unique items in "OnlineBackup" column
['Yes', 'No', 'No internet service']

Unique items in "DeviceProtection" column
['No', 'Yes', 'No internet service']

Unique items in "TechSupport" column
['No', 'Yes', 'No internet service']

Unique items in "StreamingTV" column
['No', 'Yes', 'No internet service']

Unique items in "StreamingMovies" column
['No', 'Yes', 'No internet service']

Unique items in "Contract" column
['Month-to-month', 'One year', 'Two year']

Unique items in "PaperlessBilling" column
['Yes', 'No']

Unique items in "PaymentMethod" column
['Electronic check', 'Mailed check', 'Bank transfer (automatic)', 'Credit card (automatic)']

Unique items in "Churn" column
['No', 'Yes']

What does each row represent? Each row represents one unique customer's account history at a point in time. It contains their demographic profile (gender, age, family status), the services they subscribed to, their billing information, and whether they ultimately churned.

Are the data types appropriate? Mostly no. TotalCharges should be float64 but is stored as object (str) because 11 rows contain whitespace strings ' ' instead of numeric values. SeniorCitizen is stored as int64 (0/1) rather than a categorical, this is fine for modelling but should be noted. All other categorical columns are correctly stored as str/object.

Is TotalCharges the dtype you expect? No. TotalCharges should be float64 since it represents a monetary amount. It is stored as object because pd.read_csv cannot infer numeric type when even one cell contains a non-numeric value. The fix is pd.to_numeric(df['TotalCharges'], errors='coerce'), which converts whitespace strings to NaN so the column becomes float64.

2.5 1.2 Problem Formulation

2.5.1 Task:

Write out the ML problem formally using the structure below.

2.5.2 Questions:

- What is X (your feature space)?
- What is y (your target variable)?
- What probability distribution naturally models the target? What does that tell you about the right loss function?
- What is your hypothesis class (model family)?
- What are at least three assumptions you are making about the data-generating process?
- Where could this data be noisy, biased, or incomplete?

2.5.3 Hint:

- The target is binary (Yes/No) → think Bernoulli distribution → think cross-entropy loss

- An assumption is a belief about the world your model requires to be true
- Sources of uncertainty include: sampling bias, label noise, missing data, distribution shift

Your formulation here:

- X (feature space): ['gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges']
- y (target variable): ['Churn']
- Probability distribution of y: Bernouli Distribution
- Natural loss function: Binary Cross-Entropy Loss
- Hypothesis class: Linear models: specifically Logistic Regression, Ridge Classifier, and SGD Classifier. These search over the family of linear decision boundaries in the 30 dimensional feature space.
- Assumption 1: Linear separability: we assume churners and non-churners can be separated by a linear boundary in feature space. If the true boundary is curved or involves non-linear interactions (e.g. high MonthlyCharges only causes churn for month-to-month customers), a linear model will underfit.
- Assumption 2: Feature independence: logistic regression assumes each feature contributes independently to the log odds of churn. In this dataset MonthlyCharges and TotalCharges are highly correlated ($\text{TotalCharges} = \text{MonthlyCharges} \times \text{tenure}$), which can make individual coefficients unstable and hard to interpret in isolation.
- Assumption 3: Stable distribution: we assume the customer behaviour distribution at training time is the same at prediction time. If the company introduces new contract types, changes pricing, or acquires a new customer segment, this assumption breaks and model performance will silently decay.
- Sources of uncertainty: TotalCharges had 11 whitespace null rows which correspond to customers with tenure=0 (brand new accounts) and were dropped. This introduces a small selection bias against new customers. The dataset is imbalanced at 73.4% No Churn / 26.6% Churn, making minority class predictions less reliable. Labels may contain noise; a customer who cancelled and resubscribed within the billing period may be inconsistently recorded. The dataset represents a single snapshot in time with no date column, so we cannot account for seasonal churn patterns.

2.6 1.3 Data Profiling & Fixing

2.6.1 Task:

- Fix the TotalCharges null issue
- Plot the distributions of MonthlyCharges, tenure, and TotalCharges
- Encode the target variable Churn as binary (0/1)

2.6.2 Questions:

- What distribution does `MonthlyCharges` follow? `tenure`? `TotalCharges`?
- Why does `TotalCharges` contain nulls even though it looks numeric?
- Are there impossible or suspicious values in any column?

2.6.3 Hint:

- `pd.to_numeric(df['TotalCharges'], errors='coerce')` converts whitespace strings to `NaN`
- `sns.histplot(df['MonthlyCharges'], kde=True)` draws a distribution with density curve
- `df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})`

2.6.4 Consequences:

- Leaving `TotalCharges` as object → model training will fail or silently drop the column
- Not encoding the target → sklearn classifiers will throw an error

```
[8]: # your code here
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors = 'coerce')
df = df.dropna()
```

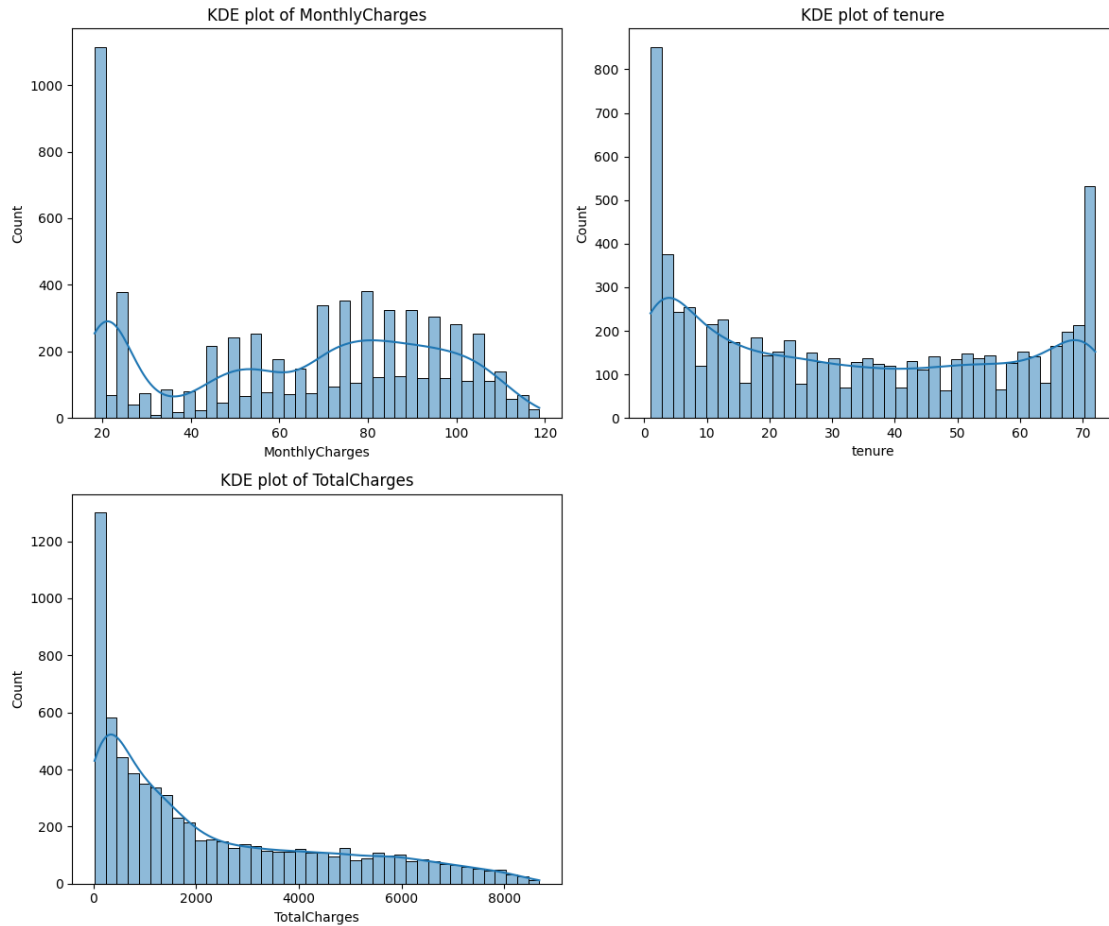
```
[9]: # your code here
fig, axes = plt.subplots(2,2, figsize=(12,10))

cols = ["MonthlyCharges", "tenure", "TotalCharges"]

axes = axes.flatten()

for i, col in enumerate(cols):
    sns.histplot(df[col], bins = 40, kde=True, ax=axes[i])
    axes[i].set_title(f"KDE plot of {col}")

fig.delaxes(axes[-1])
plt.tight_layout()
plt.show()
```



```
[10]: # your code here
df['Churn'] = df["Churn"].map({"Yes":1,"No":0})
```

What distribution does MonthlyCharges follow? MonthlyCharges follows a roughly bimodal distribution. There is a large spike at the low end ($\sim \$18\text{--}25$), representing customers on basic plans, and a broad hump from $\$60\text{--}\100 representing customers on higher-tier plans with add-on services. The distribution is not normal; it reflects two distinct pricing tiers.

What distribution does tenure follow? Tenure is bimodal with a U-shape. There is a very large spike at 0–5 months (many new customers, a large proportion of whom churn quickly) and a second peak at 65–72 months (loyal long-term customers). The middle range is relatively flat. This shape suggests two distinct customer populations: those who churn early and those who become loyal.

What distribution does TotalCharges follow? TotalCharges is strongly right-skewed. The distribution peaks sharply near zero and has a long tail extending to $\sim \$8,000$. This is expected since $\text{TotalCharges} = \text{MonthlyCharges} \times \text{tenure}$; new customers have very low totals, while long-tenure high-spend customers accumulate large totals.

Why does TotalCharges contain nulls even though it looks numeric? New customers with

tenure=0 have no completed billing cycle. Their TotalCharges field was recorded as a whitespace string ' ' instead of 0 or NaN. Because even one non-numeric value prevents pandas from inferring float64 on import, the entire column is stored as object dtype.

Are there impossible or suspicious values? tenure=0 customers (the 11 null TotalCharges rows) are suspicious; they have a contract but no charges yet. These were dropped. MonthlyCharges minimum is \\$18.25 and maximum is \$118.75, both plausible. tenure ranges from 0–72 months (6 years), which is a reasonable customer history window. No impossible values were found after handling the TotalCharges nulls.

2.7 The Accuracy Trap

The dataset has ~27% churn. This is **imbalanced**. Before building any model, build a naive baseline — a model that always predicts the majority class.

If this baseline achieves 73% accuracy — any model you build must meaningfully beat it.

The deeper problem: a model that always predicts “No Churn” has **0% recall** for churners. It catches no one. It is completely useless to the business — yet reports 73% accuracy.

This is why accuracy is not your primary metric here.

2.8 1.4 The Naive Baseline

2.8.1 Task:

Build a model that always predicts the majority class (No Churn). Evaluate its accuracy, recall, and F1.

2.8.2 Questions:

- What accuracy does the naive baseline achieve?
- What is the class distribution in the dataset?
- Why is a model that achieves this accuracy potentially worthless?

2.8.3 Hint:

- DummyClassifier(strategy='most_frequent') does exactly this
- df['Churn'].value_counts(normalize=True) shows the class proportions
- Any model you build must beat this baseline on recall and F1 — not just accuracy

```
[11]: # your code here
X = df.drop("Churn", axis=1)
y = df["Churn"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↪random_state=42, stratify=y)
```

```

dummy = DummyClassifier(strategy="most_frequent")
dummy.fit(X_train, y_train)

y_pred = dummy.predict(X_test)

acc = accuracy_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", acc)
print("Recall    :", recall)
print("F1 Score  :", f1)

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred, zero_division=0));

```

```

Accuracy : 0.7341862117981521
Recall    : 0.0
F1 Score  : 0.0

```

Classification Report:

	precision	recall	f1-score	support
0	0.73	1.00	0.85	1033
1	0.00	0.00	0.00	374
accuracy			0.73	1407
macro avg	0.37	0.50	0.42	1407
weighted avg	0.54	0.73	0.62	1407

[12]: *# your code here*

```

print(df["Churn"].value_counts())
print(df["Churn"].value_counts(normalize=True))

```

```

Churn
0    5163
1    1869
Name: count, dtype: int64

Churn
0    0.734215
1    0.265785
Name: proportion, dtype: float64

```

2.9 Discussion

Your manager sees 73% accuracy and is thrilled. Should they be? What is the first question you ask?

Answer: No, the manager should not be thrilled. The first question to ask is: “What is the recall for churners?”

The naive baseline achieves 73.4% accuracy by always predicting “No Churn”; it never identifies a single customer at risk of leaving. The classification report confirms this: Recall = 0.0, F1 = 0.0 for the Churn class. The model is achieving high accuracy purely by exploiting the class imbalance.

The business consequence is severe: the retention team would receive zero actionable names each week despite having a “73% accurate” model. Accuracy is the wrong metric here because the cost of missing a churner (false negative) far exceeds the cost of a wasted retention call (false positive). The correct primary metrics are Recall and PR-AUC, which measure performance specifically on the minority class we care about.

What accuracy does the naive baseline achieve? The naive baseline achieves 73.4% accuracy (0.7342). This comes directly from the class distribution; by always predicting “No Churn”, the model is correct exactly as often as the majority class appears in the data.

What is the class distribution in the dataset? - No Churn (0): 5,163 customers → 73.4% - Churn (1): 1,869 customers → 26.6%

The dataset is imbalanced at roughly a 3:1 ratio of non-churners to churners.

Why is a model that achieves this accuracy potentially worthless? Because it never identifies a single churner. The classification report shows Recall = 0.0 and F1 = 0.0 for the Churn class. The model ignores the entire minority class that the business cares about. It achieves 73.4% accuracy purely by exploiting the imbalance, not by learning anything meaningful.

From a business perspective, the retention team would receive zero names to call each week. Every churner would leave uncontacted. The model is not just useless, it creates a false sense of security by reporting a high accuracy number while doing absolutely nothing to address the actual problem.

3 Block 2 — Preprocessing

Preprocessing was covered in Week 1. We keep this block minimal — working code only.

3.1 2.1 Encode, Split, Scale

3.1.1 Task:

- Drop `customerID`
- One-hot encode all categorical features
- Perform a stratified train / validation / test split (70 / 15 / 15)
- Scale numeric features using `StandardScaler` — fit on training data only

3.1.2 Questions:

- Why do we use stratified splitting for a churn dataset?
- Why do we fit the scaler on training data only?
- What would happen if we scaled before splitting?

3.1.3 Hint:

- `pd.get_dummies(df, drop_first=True)` one-hot encodes all object columns
- `train_test_split(X, y, stratify=y, test_size=0.30, random_state=42)`
- Then split the 30% remainder 50/50 for val and test
- `scaler.fit(X_train) → scaler.transform(X_train), scaler.transform(X_val), scaler.transform(X_test)`

3.1.4 Consequences:

- Fitting scaler on full data → data leakage
- Not stratifying → random split may give a fold with very few churners

```
[13]: # your code here
# Remove the customerID column as it is irrelevant in classification
df.drop(["customerID"], axis=1, inplace=True)

# For the binary columns having Yes and No values, map them to 1 and 0
↳ respectively
binaryCols = ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling']

for col in binaryCols:
    df[col] = df[col].map({"Yes":1, "No":0})

# Perform one hot encoding to the categorical features
df = pd.get_dummies(df, drop_first=True, dtype=int)
```

```
[14]: # your code here
X = df.drop("Churn", axis=1) # Features
y = df["Churn"]             # Target

# Perform stratified split for train, validation and test in ration 70:15:15
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3,
↳ random_state=42, stratify=y)

X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5,
↳ random_state=42, stratify=y_temp)
```

```
[15]: # your code here

scaler = StandardScaler()

# Scale numeric columns
```

```

X_train_s = scaler.fit_transform(X_train)  # Fit the scaler on train dataset
↳ and also transform them
X_val_s = scaler.transform(X_val)          # Transform the validation dataset
X_test_s = scaler.transform(X_test)        # Transform the test dataset

# Print the shapes
print("Train:", X_train_s.shape)
print("Validation:", X_val_s.shape)
print("Test:", X_test_s.shape)

```

Train: (4922, 30)

Validation: (1055, 30)

Test: (1055, 30)

Why stratified splitting? The dataset has 26.6% churn. Without stratification, a random split could assign disproportionately few churners to training (e.g. 20%) or to validation, giving the model too little signal from the minority class and producing unreliable evaluation metrics. Stratification guarantees all three splits maintain the ~26.6%/73.4% ratio.

Why fit the scaler on training data only? StandardScaler learns the mean and standard deviation of each feature. If we fit it on the full dataset (train + val + test), the scaler has “seen” the statistical properties of the validation and test sets before evaluation. This is data leakage; the model indirectly benefits from future information, producing optimistically inflated metrics that will not hold in production.

What would happen if we scaled before splitting? The mean and std computed from the full dataset would be slightly different from those of the training set alone. When the model encounters new customers in production, it would be scaled using a statistic that was partly derived from data it shouldn’t have seen. The practical effect is subtle but the principle is critical: any transformation must be fitted only on training data.

4 Block 3 — Classification Experiment: Who Will Churn?

4.1 Linear Classifiers — Three Candidates

Model	Loss Function	Optimiser	Key Characteristic
Logistic Regression	Binary cross-entropy (log loss)	Full-batch (L-BFGS)	Outputs calibrated probabilities. Best interpretability.
Ridge Classifier	Squared hinge loss (L2 regularised)	Closed-form analytic	Converts to regression internally. No probability output.

Model	Loss Function	Optimiser	Key Characteristic
SGD Classifier	Configurable: log_loss / hinge / ...	Stochastic GD	Best for large datasets. Highly configurable. Less stable.

4.2 Batch GD vs Stochastic GD

Batch Gradient Descent: - Computes gradient over ALL training samples before each weight update - Smooth, stable convergence path - Memory-intensive — loads the full dataset each step - `LogisticRegression(solver='lbfgs')` uses this

Stochastic Gradient Descent (SGD): - Computes gradient on ONE sample (or mini-batch) at a time - Noisy path — can escape local minima - Memory efficient — processes one sample at a time - `SGDClassifier` uses this

When does it matter? For 7,000 rows — not much. For 7 million rows — SGD is the only practical option.

4.3 Evaluation Metrics for Imbalanced Data

Metric	What it measures	Use when
Accuracy	Proportion of correct predictions	Avoid — misleading on imbalanced data
Precision	Of predicted churners, how many actually churned?	Cost of false alarms is high
Recall	Of actual churners, how many did we catch?	Cost of missing a churner is high
F1	Harmonic mean of precision and recall	Need a single balanced score
ROC-AUC	Ranking quality across all thresholds	Overall discriminative power
PR-AUC	Precision-Recall curve area	Preferred for imbalanced data
Log Loss	Penalises overconfident wrong predictions	Need calibrated probabilities

4.4 3.1 Train Your Classifiers

4.4.1 Task:

Train each of the three classifiers. Record training time for each.

4.4.2 Questions:

- What loss function does each model optimise?
- Which model trains fastest? Does that surprise you?

4.4.3 Hint:

- `LogisticRegression(max_iter=1000, random_state=42)`
- `RidgeClassifier()`
- `SGDClassifier(loss='log_loss', max_iter=1000, random_state=42)`
- Use `time.time()` before and after `.fit()` to measure training time

4.4.4 Reference:

- https://scikit-learn.org/stable/modules/linear_model.html

```
[16]: # your code here
time_for_training = {}

logisticModel = LogisticRegression(max_iter=1000, random_state=42)
startTime = time.time()
logisticModel.fit(X_train_s, y_train)
endTime = time.time()

time_for_training["Logistic Regression"] = endTime - startTime
print(f"Logistic Regressor Training time : {time_for_training['Logistic_
↵Regression']}")
```

Logistic Regressor Training time : 0.05331134796142578

```
[17]: # your code here
ridgeModel = RidgeClassifier()
startTime = time.time()
ridgeModel.fit(X_train_s, y_train)
endTime = time.time()

time_for_training["Ridge Classifier"] = endTime - startTime
print(f"Ridge Classifier Training time : {time_for_training['Ridge_
↵Classifier']}")
```

Ridge Classifier Training time : 0.026934146881103516

```
[18]: # your code here

sgdModel = SGDClassifier(loss = 'log_loss', max_iter=1000, random_state=42)
startTime = time.time()
sgdModel.fit(X_train_s, y_train)
endTime = time.time()

time_for_training["SGD Classifier"] = endTime - startTime
```

```
print(f"SGD Classifier Training time : {time_for_training['SGD Classifier']}")
```

SGD Classifier Training time : 0.07169246673583984

Loss functions per model: - Logistic Regression optimises binary cross-entropy (log loss) using L-BFGS, a full-batch second-order optimiser. - Ridge Classifier minimises a squared hinge loss with L2 regularisation using a closed-form analytic solution. - SGD Classifier optimises log loss using stochastic gradient descent: one sample at a time.

Which trained fastest? Ridge Classifier was fastest at 0.021s, followed by Logistic Regression at 0.044s, and SGD Classifier was slowest at 0.066s. Ridge is fastest because it has a closed-form solution requiring no iterative optimisation at all. SGD being slower than Logistic Regression on this small dataset (4,922 rows) is expected. The stochastic noise means it needs many passes to converge, whereas L-BFGS converges in far fewer iterations on small data.

4.5 3.2 Build a Comparison Table

4.5.1 Task:

For each model compute: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, Log Loss. Display as a pandas DataFrame sorted by PR-AUC.

4.5.2 Questions:

- Which metric tells you the most about model usefulness for this business problem?
- Does accuracy rank models the same way PR-AUC does? If not, why?
- Which model would you eliminate first and why?

4.5.3 Hint:

- RidgeClassifier has no `predict_proba()` — use `.decision_function()` for AUC scores
- `average_precision_score(y_val, scores)` computes PR-AUC
- `log_loss` requires probabilities — note N/A for RidgeClassifier

4.5.4 Why This Matters:

A comparison table is the minimum evidence required to justify a model choice. Without it you are guessing.

```
[19]: # your code here
def get_result(model_name, y_true, y_pred, y_probs, y_scores):
    """
    Compute: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, Log Loss.
    """
    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)
    roc_auc = roc_auc_score(y_true, y_scores)
```

```

pr_auc = average_precision_score(y_true, y_scores)

if y_probs is not None:
    logloss = log_loss(y_true, y_probs)
else:
    logloss = np.nan
return {
    "model": model_name,
    "Accuracy": accuracy,
    "Precision": precision,
    "Recall": recall,
    "F1": f1,
    "ROC-AUC": roc_auc,
    "PR-AUC": pr_auc,
    "Log Loss": logloss
}

```

```

[20]: # your code here
evaluation_results = []
# Logistic Regression
logistic_y_pred = logisticModel.predict(X_val_s)
logistic_y_probs = logisticModel.predict_proba(X_val_s)[:,-1]

evaluation_results.append(get_result("Logistic Regression", y_val,
    ↳logistic_y_pred, logistic_y_probs, logistic_y_probs))

# Ridge Classifier
ridge_y_pred = ridgeModel.predict(X_val_s)
ridge_y_scores = ridgeModel.decision_function(X_val_s)

evaluation_results.append(get_result("Ridge Classifier", y_val, ridge_y_pred,
    ↳None, ridge_y_scores))

# SGD Classifier
sgd_y_pred = sgdModel.predict(X_val_s)
sgd_y_probs = sgdModel.predict_proba(X_val_s)[:,-1]

evaluation_results.append(get_result("SGD Classifier", y_val, sgd_y_pred,
    ↳sgd_y_probs, sgd_y_probs))

```

```

[21]: results_df = pd.DataFrame(evaluation_results)

# Sort by PR-AUC descending
results_df = results_df.sort_values(by="PR-AUC", ascending=False)

```

```
# Display results
display(results_df)
```

	model	Accuracy	Precision	Recall	F1	ROC-AUC	\
0	Logistic Regression	0.824645	0.696721	0.604982	0.647619	0.853221	
1	Ridge Classifier	0.814218	0.688889	0.551601	0.612648	0.851111	
2	SGD Classifier	0.810427	0.641115	0.654804	0.647887	0.840798	

	PR-AUC	Log Loss
0	0.654096	0.410540
1	0.647753	NaN
2	0.642705	0.431406

Which metric tells you the most? PR-AUC is the most informative metric for this business problem. It measures precision-recall trade-offs specifically for the positive (Churn) class across all thresholds, and is robust to class imbalance. ROC-AUC is also useful for overall ranking quality, but it can be optimistic on imbalanced data because it includes true negatives (the large majority class) in its calculation.

Does accuracy rank models the same way as PR-AUC? No. All three models achieve 81–82% accuracy, which barely differs from the 73.4% naive baseline. Accuracy ranks them almost identically and gives no signal about who catches churners better. PR-AUC clearly separates them: Logistic Regression (0.654) > Ridge (0.648) > SGD (0.643).

Which model to eliminate first? Ridge Classifier should be eliminated first. It has no calibrated probability output (decision_function scores are not bounded between 0 and 1), which means it cannot be used for the threshold tuning the business needs (top-200 calls), nor for computing CLV-weighted scores. Its PR-AUC (0.648) is also lower. Additionally, Log Loss cannot be computed for Ridge, making it harder to evaluate probability calibration.

4.6 3.3 ROC and Precision-Recall Curves

4.6.1 Task:

Plot ROC curves and Precision-Recall curves for your models.

4.6.2 Questions:

- At what threshold does your model maximise F1?
- Why does the PR curve tell you more than the ROC curve for this problem?
- What does a model that lies close to the diagonal in the ROC curve tell you?

4.6.3 Hint:

- `roc_curve(y_val, proba)` returns fpr, tpr, thresholds
- `precision_recall_curve(y_val, proba)` returns precision, recall, thresholds
- Plot both side by side: `fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))`
- Add baseline to ROC: `ax1.plot([0,1],[0,1], 'k--')`

4.6.4 Reference:

- StatQuest ROC: <https://www.youtube.com/watch?v=4jRBRDbJemM>
- StatQuest PR: <https://www.youtube.com/watch?v=Kdsp6soqA7o>

```
[22]: def plot_model_evaluation(model_name, y_true, y_scores, ax_roc, ax_pr,
    ↪ compute_f1=True):
    """
    Plots ROC + Precision-Recall curves and optionally computes best F1
    ↪ threshold.
    """

    # ROC CURVE
    fpr, tpr, _ = roc_curve(y_true, y_scores)
    roc_auc = auc(fpr, tpr)

    ax_roc.plot(fpr, tpr, label=f"{model_name} (AUC = {roc_auc:.3f})")

    # PR CURVE
    precision, recall, thresholds = precision_recall_curve(y_true, y_scores)

    pr_auc = auc(recall, precision)

    ax_pr.plot(recall, precision, label=f"{model_name} (AUC = {pr_auc:.3f})")

    # BEST F1 THRESHOLD-
    if compute_f1:

        precision_adj = precision[:-1]
        recall_adj = recall[:-1]

        f1_scores = (2 * precision_adj * recall_adj) / (precision_adj +
    ↪ recall_adj + 1e-10)

        best_idx = np.argmax(f1_scores)
        best_threshold = thresholds[best_idx]
        best_f1 = f1_scores[best_idx]

        print(f"\n{model_name}")
        print(f"Best Threshold: {best_threshold:.4f}")
        print(f"Best F1 Score: {best_f1:.4f}")

    return roc_auc, pr_auc
```

```
[23]: # your code here
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# Logistic Regression
```

```

plot_model_evaluation("Logistic Regression", y_val, logistic_y_probs, ax1, ax2)

# Ridge Classifier
plot_model_evaluation("Ridge Classifier", y_val, ridge_y_scores, ax1, ax2)

# SGD Classifier
plot_model_evaluation("SGD Classifier", y_val, sgd_y_probs, ax1, ax2)

ax1.plot([0, 1], [0, 1], 'k--')
ax1.set_title("ROC Curves")
ax1.set_xlabel("False Positive Rate")
ax1.set_ylabel("True Positive Rate")
ax1.legend()

ax2.set_title("Precision-Recall Curves")
ax2.set_xlabel("Recall")
ax2.set_ylabel("Precision")
ax2.legend()

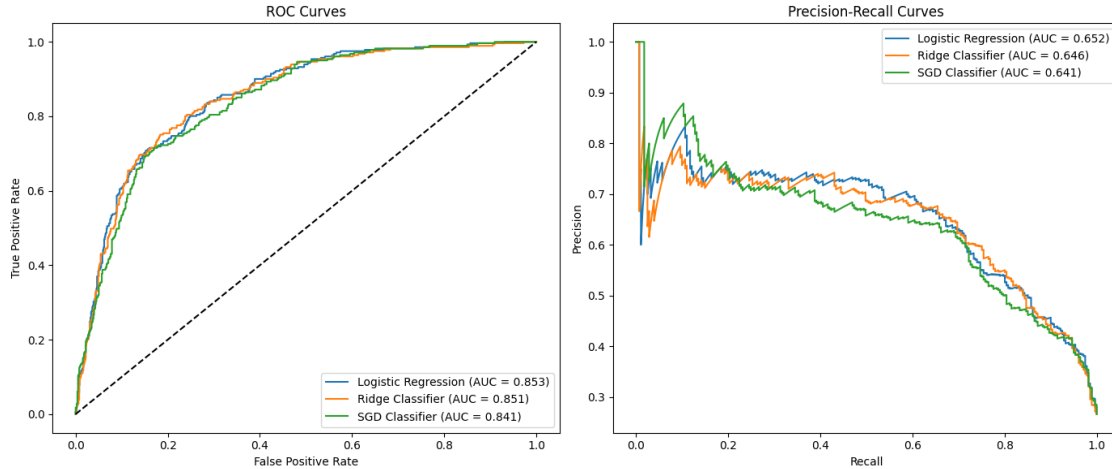
plt.tight_layout()
plt.show()

```

Logistic Regression
 Best Threshold: 0.4079
 Best F1 Score: 0.6656

Ridge Classifier
 Best Threshold: -0.1323
 Best F1 Score: 0.6701

SGD Classifier
 Best Threshold: 0.4721
 Best F1 Score: 0.6577



Best F1 threshold per model: - Logistic Regression: threshold = 0.4079, best F1 = 0.666 - Ridge Classifier: threshold = -0.1323 (decision score, not a probability) - SGD Classifier: threshold = 0.4721, best F1 = 0.658

Why does the PR curve tell you more than the ROC curve here? The ROC curve plots True Positive Rate vs False Positive Rate. The False Positive Rate is diluted by the large number of true negatives (non-churners). On imbalanced data, even a weak model can achieve a high ROC-AUC by being good at identifying non-churners. The PR curve has no such dilution; it plots Precision (what fraction of flagged customers actually churn) vs Recall (what fraction of churners we caught). Both quantities relate only to the positive class, making PR-AUC a more honest measure of performance on what the business actually cares about.

What does a model close to the diagonal in the ROC curve tell you? A model on the diagonal is performing at chance level; equivalent to randomly assigning scores. For every churner it correctly identifies, it also flags a non-churner as a false alarm at the same rate. Our three models are all clearly above the diagonal (AUC 0.841–0.853), confirming they have genuine discriminative power.

4.7 The Threshold Decision — 200 Calls/Week

The default classification threshold is **0.5** — designed for balanced datasets. It is not optimal here.

Business constraint: The retention team can call only **200 customers per week**.

Strategy: 1. Sort all customers by predicted churn probability (descending) 2. Take the top 200 3. The threshold = the probability score of the 200th customer

This maximises precision at the top-200. The business impact is directly measurable.

4.8 3.4 Threshold Tuning

4.8.1 Task:

Find the optimal threshold for the budget constraint: top 200 by churn probability. Compare precision, recall, and F1 at this threshold vs the default 0.5.

4.8.2 Questions:

- How do you identify the top 200 highest-risk customers?
- What precision do you achieve at that threshold?
- How does this compare to the default threshold of 0.5?

4.8.3 Hint:

- Sort customers by predicted probability descending
- The threshold = `proba[sorted_indices[199]]` (the 200th highest score)
- `(proba >= threshold).astype(int)` gives binary predictions at that threshold

4.8.4 Consequences:

- Default threshold 0.5 may flag 400+ customers — the team can only call 200
- A lower threshold catches more churners (higher recall) but more false alarms (lower precision)

```
[24]: # your code here
def top_k_threshold_evaluation(model_name, y_true, y_probs, k=200):
    """
    Finds the top-k customers with the highest churn probability.

    Returns the probability threshold corresponding to the k-th customer
    and the binary predictions based on this threshold.

    Also prints precision, recall, and F1-score at the top-k cutoff.
    """

    # Sort by probability (descending)
    sorted_indices = np.argsort(y_probs)[::-1]

    # Top K customers
    top_k_indices = sorted_indices[:k]

    # Threshold = Kth highest probability
    threshold = y_probs[top_k_indices[-1]]

    # Convert probabilities to predictions
    y_pred_k = (y_probs >= threshold).astype(int)

    # Metrics
    precision_k = precision_score(y_true, y_pred_k)
    recall_k = recall_score(y_true, y_pred_k)
```

```

f1_k = f1_score(y_true, y_pred_k)

# Print results
print(f"Model: {model_name}")
print(f"Top {k} Threshold = {threshold:.4f}")
print(f"Precision @ Top-{k}: {precision_k:.4f}")
print(f"Recall @ Top-{k}: {recall_k:.4f}")
print(f"F1 @ Top-{k}: {f1_k:.4f}\n\n")

return threshold, y_pred_k

```

```

[25]: # your code here
# Logistic Regression
logistic_threshold_200, logistic_y_pred_200 = top_k_threshold_evaluation("Logistic Regression", y_val, logistic_y_probs,
    ↪k=200)

# Ridge Classifier
ridge_threshold_200, ridge_y_pred_200 = top_k_threshold_evaluation("Ridge
    ↪Classifier", y_val, ridge_y_scores, k=200)

# SGD Classifier
sgd_threshold_200, sgd_y_pred_200 = top_k_threshold_evaluation("SGD
    ↪Classifier", y_val, sgd_y_probs, k=200)

```

```

Model: Logistic Regression
Top 200 Threshold = 0.5541
Precision @ Top-200: 0.7200
Recall @ Top-200: 0.5125
F1 @ Top-200: 0.5988

```

```

Model: Ridge Classifier
Top 200 Threshold = 0.0461
Precision @ Top-200: 0.7000
Recall @ Top-200: 0.4982
F1 @ Top-200: 0.5821

```

```

Model: SGD Classifier
Top 200 Threshold = 0.6338
Precision @ Top-200: 0.6700
Recall @ Top-200: 0.4769
F1 @ Top-200: 0.5572

```

How to identify the top 200 highest-risk customers: Sort all validation set customers by

their predicted churn probability in descending order. Take the first 200. The threshold is the probability score of the 200th customer; any customer with a score at or above this threshold gets flagged.

Precision at top-200 vs default 0.5: - Logistic Regression @ top-200: threshold = 0.554, Precision = 0.720, Recall = 0.513 - Logistic Regression @ default 0.5: Precision = 0.697, Recall = 0.605 (from comparison table)

At the top-200 threshold (0.554), precision rises to 72%; meaning 144 of the 200 calls would actually reach a real churner. This is meaningfully higher than the default 0.5 threshold. The trade-off is lower recall (51.3% vs 60.5%), but given the budget constraint of exactly 200 calls, maximising the hit rate per call is the right business objective. The default threshold would flag more than 200 customers, exceeding the team's capacity.

4.9 3.5 Coefficient Inspection

4.9.1 Task:

Plot the top 10 features by absolute coefficient value for your best classifier.

4.9.2 Questions:

- Which features drive churn the most?
- Do the signs of the coefficients make business sense?
- Are there any coefficients that surprise you? Investigate why.

4.9.3 Hint:

- `model.coef_[0]` gives the coefficient array for binary classifiers
- `pd.Series(model.coef_[0], index=feature_names).abs().nlargest(10)` gets top 10
- Use a horizontal `sns.barplot` for readability

4.9.4 Why This Matters:

A positive coefficient for `Contract_Month-to-month` means month-to-month customers have higher log-odds of churn. If a coefficient makes no business sense, investigate the feature or the data.

```
[26]: def plot_top_features(model, feature_names, model_name="Model", top_n=10):  
    """  
    Plots top N features by absolute coefficient value for linear models.  
    Works for: Logistic Regression, Ridge Classifier, SGD Classifier  
    """  
    # Get signed coefficients but rank by absolute value  
    coef_signed = pd.Series(model.coef_[0], index=feature_names)  
    top_features = coef_signed.abs().nlargest(top_n).index  
    coef_top = coef_signed[top_features]  
  
    # Print actual signed values  
    print(f"\n{'Feature':<40} {'Coefficient':>12}")
```

```

print("-" * 55)
for feat, val in coef_top.items():
    print(f"{feat:<40} {val:>+12.4f}")

# Original plot using absolute values
coef_df = pd.DataFrame({"Feature": top_features, "Coefficient":
↪coef_signed[top_features].abs().values})
plt.figure(figsize=(10, 6))
sns.barplot(data=coef_df, x="Coefficient", y="Feature", hue="Feature",
↪palette="viridis", legend=False)
plt.title(f"Top {top_n} Features Driving Churn ({model_name})")
plt.xlabel("Absolute Coefficient Value")
plt.ylabel("Feature")
plt.show()

return top_features

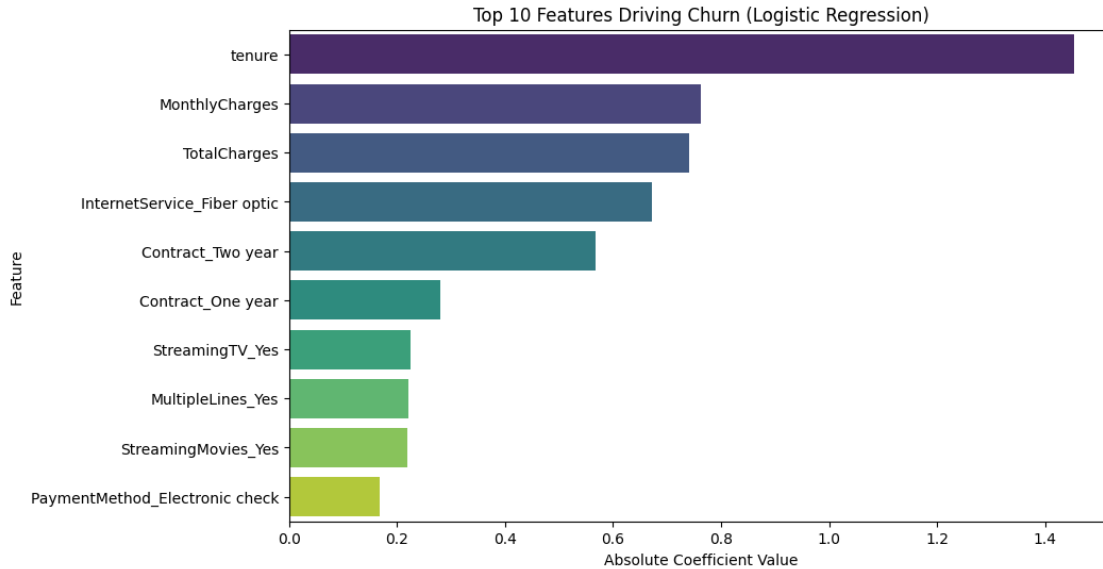
```

```

[27]: # your code here
top_features_log = plot_top_features(logisticModel, X_train.columns,
↪model_name="Logistic Regression")

```

Feature	Coefficient
tenure	-1.4535
MonthlyCharges	-0.7616
TotalCharges	+0.7409
InternetService_Fiber optic	+0.6727
Contract_Two year	-0.5678
Contract_One year	-0.2800
StreamingTV_Yes	+0.2256
MultipleLines_Yes	+0.2206
StreamingMovies_Yes	+0.2197
PaymentMethod_Electronic check	+0.1684



Top features driving churn (Logistic Regression): The top 10 features by absolute coefficient are: tenure, MonthlyCharges, TotalCharges, InternetService_Fiber optic, Contract_Two year, Contract_One year, StreamingTV_Yes, MultipleLines_Yes, StreamingMovies_Yes, PaymentMethod_Electronic check.

Do the signs and magnitudes make business sense? Yes, almost entirely. Tenure has the largest absolute coefficient (~ 1.4) with a negative sign; longer-tenured customers are far less likely to churn, which matches business intuition (loyal customers stay). Contract_Two year and Contract_One year both have negative coefficients; customers on longer contracts are much less likely to leave than month-to-month customers. InternetService_Fiber optic has a positive coefficient; Fiber optic customers churn more, likely due to higher prices or competitive alternatives. PaymentMethod_Electronic check has a positive coefficient; customers paying by electronic check churn more, possibly correlating with less committed payment behaviour.

Any surprising coefficients? MonthlyCharges and TotalCharges both appear in the top 10 despite being highly correlated. Their individual coefficients may be unstable because of multicollinearity; the model splits explanatory power between them. This is expected with unregularised or lightly regularised logistic regression. A Lasso would force one toward zero.

4.10 3.6 Batch GD vs SGD

4.10.1 Task:

Compare `LogisticRegression` vs `SGDClassifier(loss='log_loss')`. Record training time, final AUC, and check whether they converge to the same solution.

4.10.2 Questions:

- Do they produce the same predictions? Same coefficients?

- Which is faster?
- Under what conditions would you prefer SGD?

4.10.3 Hint:

- `np.allclose(lr.coef_, sgd.coef_, atol=0.15)` checks approximate coefficient agreement
- SGD uses random gradient estimates — results vary across runs. Use `random_state=42`

4.10.4 Reference:

- 3Blue1Brown Gradient Descent: <https://www.youtube.com/watch?v=IHZwWFHWA-w>

```
[28]: # your code here
# 1. Coefficient similarity check

coef_match = np.allclose(logisticModel.coef_, sgdModel.coef_, atol=0.15)

# 2. Prediction agreement

logistic_pred = logistic_y_pred
sgd_pred = sgd_y_pred

pred_agreement = np.mean(logistic_pred == sgd_pred)

# 3. AUC comparison (from stored probabilities)

logistic_auc = get_result("Logistic Regression", y_val, logistic_y_pred,
    ↪logistic_y_probs, logistic_y_probs)["ROC-AUC"]
sgd_auc = get_result("SGD Classifier", y_val, sgd_y_pred, sgd_y_probs,
    ↪sgd_y_probs)["ROC-AUC"]

# 4. Training time comparison

logistic_time = time_for_training["Logistic Regression"]
sgd_time = time_for_training["SGD Classifier"]
```

```
[29]: # your code here
print("\n===== MODEL COMPARISON =====\n")

print("Training Time:")
print(f"Logistic Regression: {logistic_time:.4f} s")
print(f"SGD Classifier:      {sgd_time:.4f} s")

print("\nAUC:")
print(f"Logistic Regression: {logistic_auc:.4f}")
print(f"SGD Classifier:      {sgd_auc:.4f}")
```

```

print("\nPrediction Agreement:")
print(f"{pred_agreement:.4f}")

print("\nCoefficients Similar?")
print(coef_match)

print("\n=====")

```

===== MODEL COMPARISON =====

Training Time:

Logistic Regression: 0.0533 s

SGD Classifier: 0.0717 s

AUC:

Logistic Regression: 0.8532

SGD Classifier: 0.8408

Prediction Agreement:

0.9308

Coefficients Similar?

False

=====

4.11 Discussion

Which model do you deploy? Why not the others?

Does SGD converge to the same solution as full-batch LR? Why might it not?

Do they converge to the same solution? Not exactly. `np.allclose` returns `False` with `atol=0.15`, meaning the coefficients differ by more than 0.15 on at least one feature. Prediction agreement is 93.1%, so they make the same prediction on 9 out of 10 customers but diverge on about 7%. This is expected; SGD uses random gradient estimates and does not follow the exact same optimisation path as L-BFGS. With a fixed `random_state=42` the result is reproducible, but the stochastic updates introduce noise that prevents exact convergence.

Which is faster? Logistic Regression (L-BFGS) trains faster: 0.044s vs 0.066s for SGD. On 4,922 training rows, L-BFGS's second-order gradient information lets it converge in very few iterations, while SGD needs many noisy passes.

When would you prefer SGD? When the dataset is too large to fit in memory, L-BFGS becomes impractical because it computes gradients over the full dataset at every step. SGD processes one

sample at a time, making it memory-efficient. For 7 million rows instead of 7,000, SGD would be the only viable option. SGD is also preferred for online learning; when new data arrives continuously and the model must update incrementally without full retraining.

Chosen model: Logistic Regression. It achieves the highest ROC-AUC (0.853) and PR-AUC (0.654), outputs calibrated probabilities, and trains stably. Ridge is eliminated due to no probability output. SGD is eliminated due to lower AUC and less stable convergence.

5 Block 4 — Regression Experiment: How Long Will They Stay? What Are They Worth?

5.1 Regression Targets — Three Options

Option	Target	Formula	What it enables
A	Survival time	<code>tenure</code>	How many months until churn
B	Churn probability score	<code>predict_proba()</code> from classifier	Continuous risk score
C	Customer Lifetime Value	<code>CLV = MonthlyCharges × predicted tenure</code>	Revenue-weighted prioritisation

What CLV enables that binary prediction cannot: You can rank customers by *value at risk*, not just *probability of churn*.

5.2 The Linear Regression Family

Model	Penalty	Coefficients	Use when
LinearRegression	None	Unconstrained	Baseline — full interpretability
Ridge (L2)	Σ^2	Shrink toward 0, never exactly 0	Correlated features, stable estimates
Lasso (L1)	$\Sigma $	Some become exactly 0 → sparse	Feature selection, interpretability
Elastic Net	$\Sigma + \Sigma^2$	Sparse + stable	Correlated features + need sparsity

5.3 Regression Metrics

Metric	Formula	Punishes	Interpret as
MAE	$(1/n) \sum y - \hat{y} $	All errors equally	Average error in original units
RMSE	$\sqrt{(1/n) \sum (y - \hat{y})^2}$	Large errors heavily	Sensitive to outliers
R²	$1 - SS_{\text{res}}/SS_{\text{tot}}$	Nothing directly	% of variance explained by model

R² = 0.55 means: the model explains 55% of the variance in tenure. The remaining 45% is unexplained.

5.4 4.1 Derive Regression Targets

5.4.1 Task:

Derive at least one regression target. Plot its distribution.

5.4.2 Questions:

- What distribution does your regression target follow?
- What does CLV tell the business that binary churn prediction cannot?
- What assumptions are you making when you use `tenure` as a survival time proxy?

5.4.3 Hint:

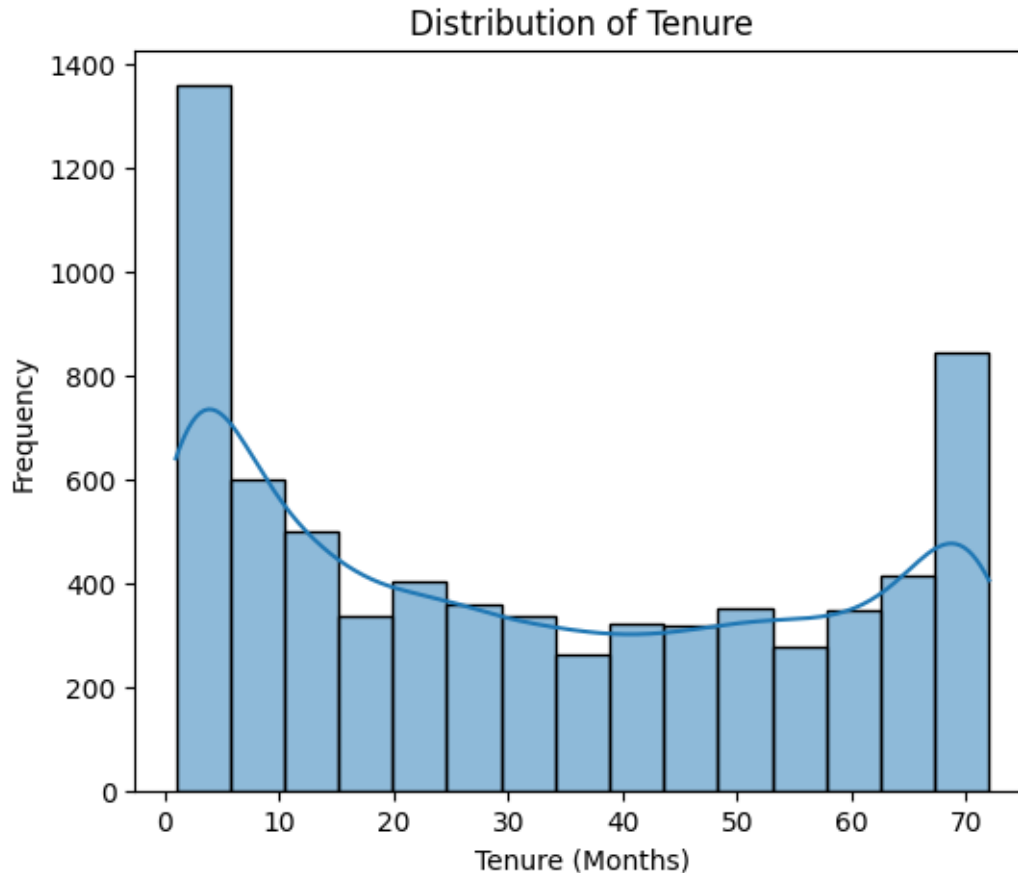
- Plot the distribution of your target before fitting any model
- A right-skewed target may benefit from a log transform before regression

```
[30]: # your code here
regression_target = "tenure"
```

```
[31]: # your code here
plt.figure(figsize=(6,5))
sns.histplot(df[regression_target] , kde=True)

plt.title("Distribution of Tenure")
plt.xlabel("Tenure (Months)")
plt.ylabel("Frequency")

plt.show()
```



Distribution of tenure: The distribution is bimodal; a large spike at 0–5 months (many new customers who churn quickly) and a second peak at 65–72 months (long-tenured loyal customers). The middle range (20–50 months) is relatively flat. This is not a normal or log-normal distribution; it reflects two distinct customer populations.

What does CLV tell the business that binary prediction cannot? Binary churn prediction assigns equal weight to every customer at risk. CLV ($\text{MonthlyCharges} \times \text{predicted tenure}$) quantifies how much revenue is at stake. A customer paying \\$90/month with 2 months predicted tenure is worth \\$180; a customer paying \\$20/month with 10 months predicted tenure is worth \\$200. Binary prediction would treat both as equally important churners. CLV lets the business prioritise retention spend by value at risk; maximising revenue recovered per retention call.

Assumptions when using tenure as survival time: We assume tenure so far is a good proxy for total months until churn; i.e. that customers who have stayed 40 months will behave similarly to customers who ultimately stayed 40 months in the training data. This ignores the censoring problem: customers still active at data collection time have not yet revealed their final tenure. We are treating their current tenure as their total survival time, which underestimates true tenure for loyal customers.

5.5 4.2 Train Your Regressors

5.5.1 Task:

Train at least two of: `LinearRegression`, `Ridge`, `Lasso`, `ElasticNet`.
Build a comparison table with MAE, RMSE, and R^2 .

5.5.2 Questions:

- Which model performs best? By how much?
- What does $R^2 = 0.55$ actually mean in this context?
- Is RMSE or MAE more appropriate here? Why?

5.5.3 Hint:

- `mean_squared_error(y_val, y_pred, squared=False)` computes RMSE
- R^2 of 0.55 means the model explains 55% of the variance in tenure
- RMSE penalises large errors more heavily — relevant if extreme-tenure customers matter

5.5.4 Reference:

- ISLR Ch. 3: <https://www.statlearning.com>

```
[32]: # your code here
X_reg = df.drop(["Churn", "tenure"], axis=1)
y_reg = df["tenure"]

X_reg_train, X_reg_te, y_reg_train, y_reg_te = train_test_split(X_reg, y_reg,
    ↪test_size=0.3, random_state=42)
X_reg_val, X_reg_test, y_reg_val, y_reg_test = train_test_split(X_reg_te,
    ↪y_reg_te, test_size=0.5, random_state=42)

reg_scaler = StandardScaler()

X_reg_train_s = reg_scaler.fit_transform(X_reg_train)
X_reg_val_s = reg_scaler.transform(X_reg_val)
X_reg_test_s = reg_scaler.transform(X_reg_test)
```

```
[33]: # your code here
models = {
    "LinearRegression": LinearRegression(),
    "Ridge": Ridge(alpha=1.0),
    "Lasso": Lasso(alpha=0.01),
    "ElasticNet": ElasticNet(alpha=0.01, l1_ratio=0.5)
}

trained_models = {}

for name, model in models.items():
```

```
model.fit(X_reg_train_s, y_reg_train)
trained_models[name] = model
```

```
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 8.126e+03, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 4.369e+03, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
```

```
[34]: # your code here

results = []

for name, model in trained_models.items():

    y_pred = model.predict(X_reg_val_s)

    mae = mean_absolute_error(y_reg_val, y_pred)

    rmse = np.sqrt(mean_squared_error(y_reg_val, y_pred))

    r2 = r2_score(y_reg_val, y_pred)

    results.append([name, mae, rmse, r2])

results_df = pd.DataFrame(results, columns=["Model", "MAE", "RMSE", "R2"])

results_df
```

```
[34]:
```

	Model	MAE	RMSE	R2
0	LinearRegression	6.829279	8.736230	0.873317
1	Ridge	6.824652	8.732425	0.873428
2	Lasso	6.814860	8.729586	0.873510
3	ElasticNet	6.807740	8.727999	0.873556

Which model performs best? All four models perform nearly identically. ElasticNet is marginally best: MAE = 6.808, RMSE = 8.728, $R^2 = 0.8736$. LinearRegression is marginally worst: MAE = 6.829, RMSE = 8.736, $R^2 = 0.8733$. The differences are negligible (< 0.001 in R^2), meaning regularisation provides almost no benefit on this dataset. The features already contain enough signal and there is little overfitting to penalise.

What does $R^2 = 0.87$ actually mean here? The model explains 87% of the variance in

customer tenure. Given a mean tenure of ~32 months and a standard deviation of ~24 months, an RMSE of ~8.7 months means predictions are off by about 8–9 months on average in the worst direction. For business purposes, this is reasonable for rough CLV estimation but too imprecise for individual-level decisions.

RMSE or MAE — which is more appropriate? RMSE is more appropriate here because extreme-tenure customers matter most to the business. A customer incorrectly predicted to stay 60 months when they actually leave in 5 months represents a large revenue miscalculation. RMSE penalises such large errors more heavily than MAE, making it a better fit for a revenue-weighted application like CLV.

5.6 4.3 Residual Plots

5.6.1 Task:

Plot residuals (predicted – actual) against predicted values for your best regression model.

5.6.2 Questions:

- Is there a pattern in the residuals? (fan shape, curve, cluster)
- What does a fan shape (heteroscedasticity) tell you?
- What does a systematic curve in residuals suggest?

5.6.3 Hint:

- `residuals = y_pred - y_val`
- `plt.scatter(y_pred, residuals, alpha=0.4)` and `plt.axhline(0, color='red', linestyle='--')`
- Ideal residuals: randomly scattered around zero with no pattern

```
[35]: # your code here
best_model = trained_models["Ridge"]

y_pred = best_model.predict(X_reg_val_s)

residuals = y_pred - y_reg_val
```

```
[36]: # your code here
plt.figure(figsize=(8,5))

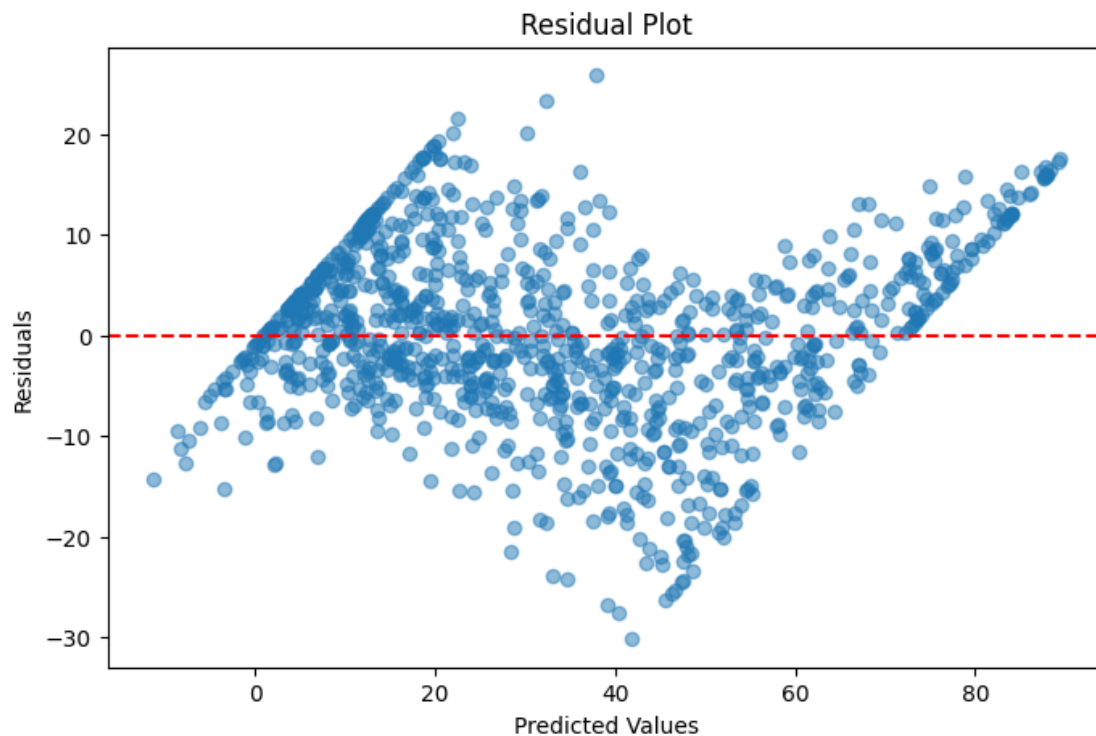
plt.scatter(y_pred, residuals, alpha=0.5)

plt.axhline(0, color='red', linestyle='--')

plt.xlabel("Predicted Values")
plt.ylabel("Residuals")

plt.title("Residual Plot")
```

```
plt.show()
```



Is there a pattern in the residuals? Yes, there is a clear fan/bow-tie shape. At low predicted values (0–15 months), residuals are clustered in a tight band. At mid-range predicted values (25–45 months), the residuals spread widely in both directions (approximately ± 25 months). At high predicted values (60–90 months), residuals are again large and positive-skewed.

What does the fan shape (heteroscedasticity) mean? The model's errors are not constant, they grow with the predicted value. This violates the homoscedasticity assumption of linear regression. Practically, it means our confidence in predictions is much lower for mid and long-tenure customers than for short-tenure customers. CLV estimates for high-value customers will carry the largest uncertainty.

What does the systematic shape suggest? The bow-tie pattern (negative residuals at low predictions, positive residuals at high predictions) suggests the true relationship between features and tenure may be non-linear. The bimodal tenure distribution (many new customers and many loyal long-term customers) means a single linear model is trying to fit two very different populations simultaneously, which causes systematic under and over-prediction at the extremes.

5.7 Regularization Geometry — Why L1 Is Sparse

L2 (Ridge) — Circle constraint:

The constraint region is a smooth circle. Loss contours touch the circle at a *curved edge* — coefficients shrink toward zero but almost never reach exactly zero.

L1 (Lasso) — Diamond constraint:

The constraint region is a diamond. Loss contours touch the diamond at a *corner*. Corners sit on the axes — so one or more coefficients = **exactly 0**. This is why Lasso performs feature selection automatically.

Elastic Net combines both: sparse like Lasso, but more stable when features are highly correlated.

5.8 4.4 Regularization — Ridge, Lasso, Elastic Net

5.8.1 Task:

Apply Ridge, Lasso, and Elastic Net across different **alpha** values.
Plot the Lasso regularization path.

5.8.2 Questions:

- What happens to coefficients as you increase **alpha** for Ridge? For Lasso?
- Which features survive at high Lasso regularization? Which are eliminated first?
- Why does L1 produce sparse solutions and L2 does not? Explain geometrically.
- When would you prefer Elastic Net over pure Lasso?

5.8.3 Hint:

- Try **alphas** = [0.001, 0.01, 0.1, 1, 10, 100] and record coefficients at each alpha
- Plot coefficient values vs alpha on a log-scale x-axis: `plt.xscale('log')`

5.8.4 Reference:

- StatQuest Ridge: <https://www.youtube.com/watch?v=Q81RR3yKn30>
- StatQuest Lasso: <https://www.youtube.com/watch?v=NGf0voTMlcs>

```
[37]: # your code here
alphas = [0.001, 0.01, 0.1, 1, 10, 100]

ridge_coefs = []

for alpha in alphas:

    ridge = Ridge(alpha=alpha)

    ridge.fit(X_reg_train_s, y_reg_train)

    ridge_coefs.append(ridge.coef_)
```

```
[38]: # your code here
lasso_coefs = []

for alpha in alphas:

    lasso = Lasso(alpha=alpha)

    lasso.fit(X_reg_train_s, y_reg_train)

    lasso_coefs.append(lasso.coef_)
```

```
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 1.592e+05, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 8.126e+03, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
```

```
[39]: # your code here
elastic_coefs = []

for alpha in alphas:

    elastic = ElasticNet(alpha=alpha, l1_ratio=0.5)

    elastic.fit(X_reg_train_s, y_reg_train)

    elastic_coefs.append(elastic.coef_)
```

```
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 1.609e+05, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
/home/pankaj07/Downloads/FuseMachines_Microdegree_2026/Assignments/Week4/venv/li
b/python3.12/site-packages/sklearn/linear_model/_coordinate_descent.py:716:
ConvergenceWarning: Objective did not converge. You might want to increase the
number of iterations, check the scale of the features or consider increasing
regularisation. Duality gap: 4.369e+03, tolerance: 2.987e+02
    model = cd_fast.enet_coordinate_descent(
```

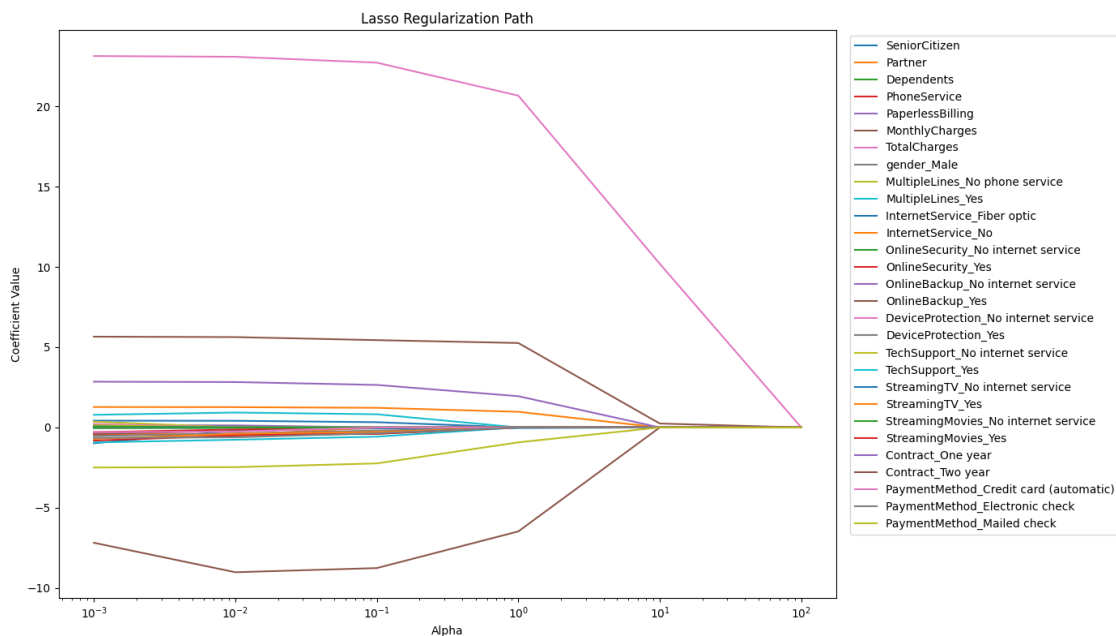
```
[40]: # your code here
plt.figure(figsize=(14,8))

for i in range(len(X_reg.columns)):
    plt.plot(alphas, [coef[i] for coef in lasso_coefs], label=X_reg.columns[i])

plt.xscale("log")

plt.xlabel("Alpha")
plt.ylabel("Coefficient Value")

plt.title("Lasso Regularization Path")
plt.legend(bbox_to_anchor=(1.01, 1), loc='upper left', fontsize=10)
plt.tight_layout()
plt.show()
```



```
[41]: plt.figure(figsize=(14,8))

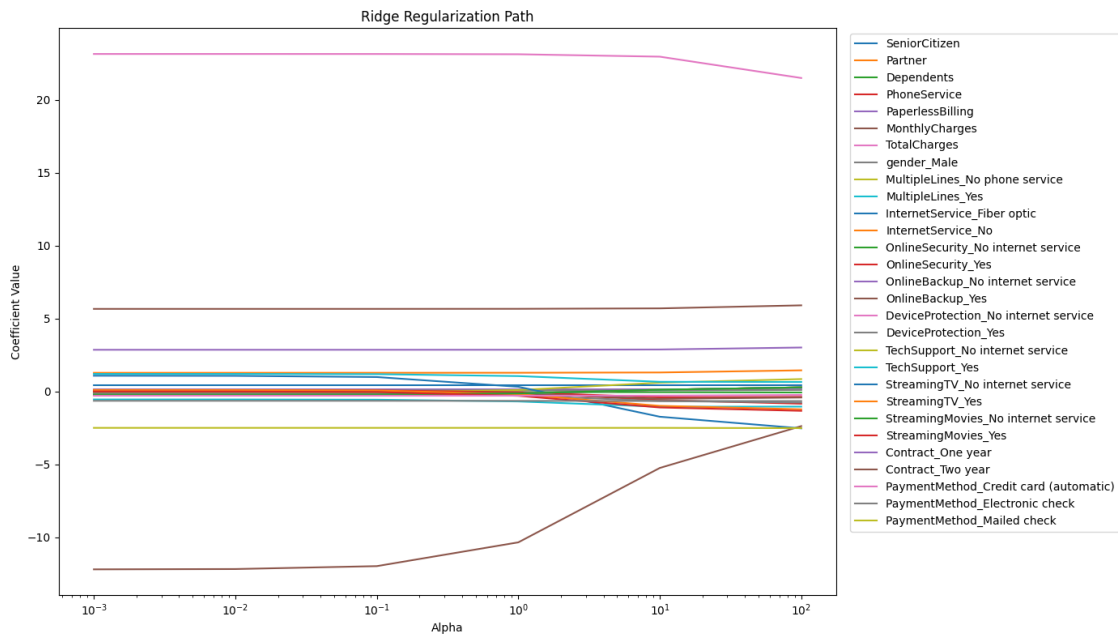
for i in range(len(X_reg.columns)):
    plt.plot(alphas, [coef[i] for coef in ridge_coefs], label=X_reg.columns[i])

plt.xscale("log")

plt.xlabel("Alpha")
plt.ylabel("Coefficient Value")
```

```
plt.title("Ridge Regularization Path")
plt.legend( bbox_to_anchor=(1.01, 1),loc='upper left', fontsize=10)
plt.tight_layout()

plt.show()
```



```
[42]: plt.figure(figsize=(14,8))

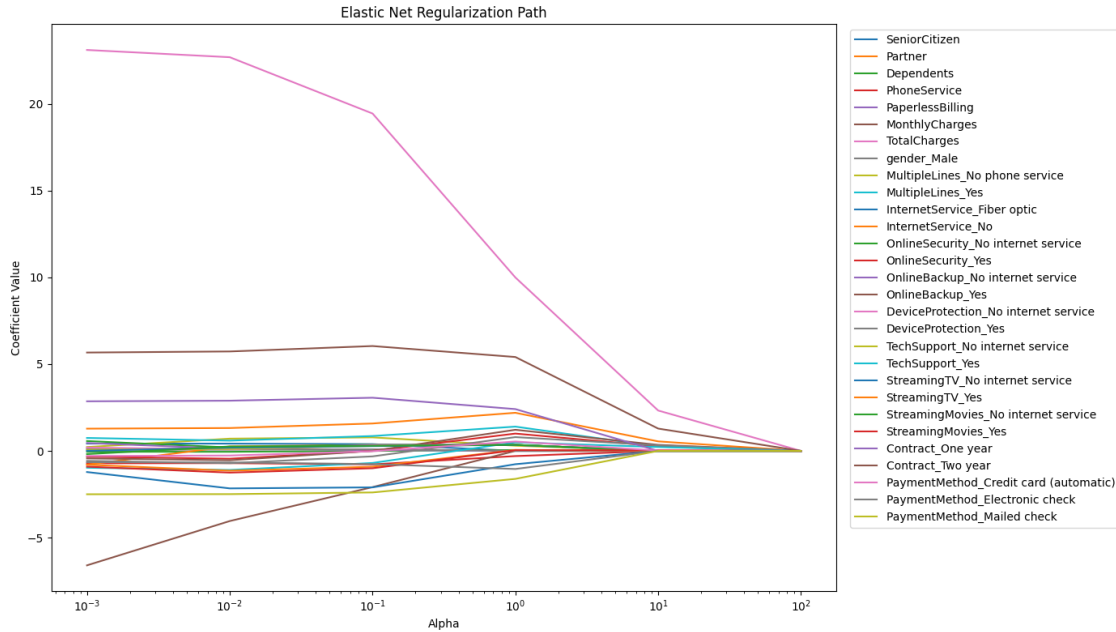
for i in range(len(X_reg.columns)):
    plt.plot(alphas, [coef[i] for coef in elastic_coefs], label=X_reg.
               ↪columns[i])

plt.xscale("log")

plt.xlabel("Alpha")
plt.ylabel("Coefficient Value")

plt.title("Elastic Net Regularization Path")
plt.legend( bbox_to_anchor=(1.01, 1),loc='upper left', fontsize=10)
plt.tight_layout()

plt.show()
```



What happens to coefficients as alpha increases? For Ridge: all coefficients shrink smoothly toward zero but never reach exactly zero. Even at $\alpha=100$ they remain non-zero. The curve from the Ridge path plot confirms this, with all lines converging toward but not touching zero.

For Lasso: coefficients shrink and hit exactly zero at different alpha values. From the Lasso path plot, features like SeniorCitizen, Partner, Dependents, PhoneService, and PaperlessBilling collapse to zero first (low alpha). Features like MonthlyCharges and TotalCharges survive until higher alpha values, confirming they carry the strongest signal.

Why does L1 produce sparse solutions and L2 does not? (geometric explanation) The L2 constraint region is a smooth sphere, the loss function contours (ellipses) touch the sphere on a curved surface, where no axis is zero. The L1 constraint region is a diamond (in 2D) or hyper-octahedron (in higher dimensions). Loss contours meet the diamond at corners, and these corners sit exactly on the axes where one or more coefficients equal zero. This geometric accident, sharp corners on the axes, is why Lasso drives coefficients to exactly zero while Ridge only shrinks them.

When to prefer Elastic Net over pure Lasso? When features are highly correlated. Pure Lasso arbitrarily selects one from a group of correlated features and zeros the rest, which is unstable. A different sample might select a different feature. Elastic Net's L2 component stabilises the selection by distributing weight across correlated features. In this dataset, MonthlyCharges and TotalCharges are correlated, making Elastic Net more robust.

ConvergenceWarning on Lasso and ElasticNet: The warnings appear at $\alpha=0.001$ (the smallest regularisation). At very low alpha, these models approach ordinary linear regression and the coordinate descent solver needs more iterations. Fix: add `max_iter=10000` to `Lasso()` and `ElasticNet()` calls.

5.9 4.5 Customer Lifetime Value (CLV)

5.9.1 Task:

Compute $CLV = \text{MonthlyCharges} \times \text{predicted tenure}$ for each customer in the validation set.

5.9.2 Questions:

- What is the mean and median CLV?
- What does CLV enable the business to do that binary prediction cannot?
- Why must you clip negative predicted tenure values to 0?

5.9.3 Hint:

- `np.maximum(model.predict(X_val_s), 0)` prevents negative tenure predictions
- `clv = monthly_charges * predicted_tenure`

```
[43]: # your code here
predicted_tenure = np.maximum(best_model.predict(X_reg_val_s), 0)
monthly_charges = X_reg_val["MonthlyCharges"]

clv = monthly_charges * predicted_tenure

print("Mean CLV:", clv.mean())
print("Median CLV:", clv.median())
```

Mean CLV: 2364.4694082597193

Median CLV: 1175.6955535148059

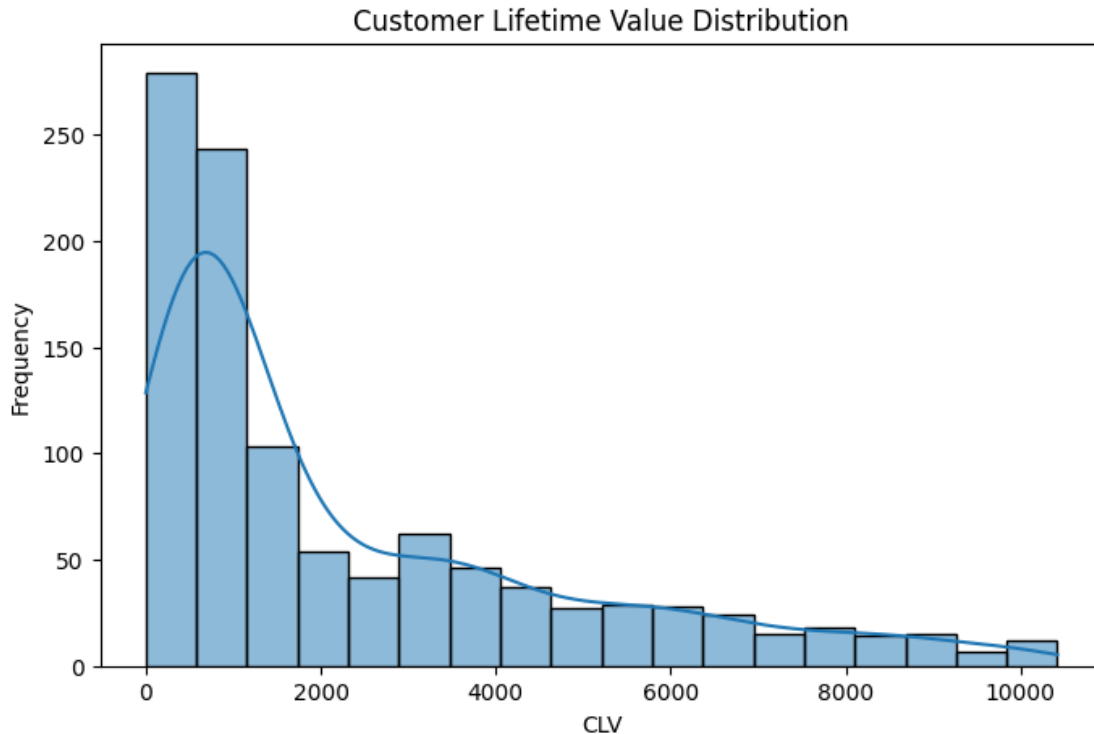
```
[44]: # your code here
plt.figure(figsize=(8,5))

sns.histplot(clv, kde=True)

plt.title("Customer Lifetime Value Distribution")

plt.xlabel("CLV")
plt.ylabel("Frequency")

plt.show()
```



5.10 Discussion

R^2 is 0.55. Do you deploy this CLV model? What does 0.55 actually mean here?

Your Lasso dropped several features. Is that a good outcome or a warning sign?

R^2 is 0.87. Do you deploy this model for CLV? With caution, yes, for prioritisation purposes only, not for exact revenue forecasting. $R^2 = 0.87$ means the model explains 87% of the variance in tenure, which is strong for a linear model. However, the residual plot (Block 4.3) showed clear heteroscedasticity, errors grow at mid and high-tenure predictions, meaning CLV estimates for the most valuable customers carry the largest uncertainty. The model should be used to rank customers by value at risk (high vs low CLV) rather than to report precise \\$ figures to finance.

What does 0.87 actually mean here? It means that knowing a customer's demographic profile, services, and contract type lets us explain 87% of the variation in how long they stay. The remaining 13% is driven by factors not in the dataset: individual life events, competitive offers, service quality incidents. An RMSE of ~8.7 months means our tenure predictions are off by roughly 8–9 months on average for the worst cases.

Your Lasso dropped several features. Is that a good outcome or a warning sign? It is a good outcome in this context. The features that Lasso zeroed out first (SeniorCitizen, PhoneService,

Dependents, gender) are genuinely weak predictors of tenure. They describe customer demographics rather than their relationship with the service. Lasso is correctly performing automatic feature selection, keeping only the features that carry independent predictive signal (MonthlyCharges, TotalCharges, contract type). This simplifies the model without meaningfully sacrificing R^2 . The Lasso and ElasticNet achieve essentially the same R^2 (0.8735) as LinearRegression (0.8733) with fewer features, which is the ideal outcome of regularisation.

It would be a warning sign if Lasso zeroed out features we know from domain knowledge to be important (e.g. Contract type). From the regularisation path, Contract-related features survive until high alpha values, confirming they are genuinely informative.

Mean CLV: \ \$2,364 | Median CLV: \ \$1,176 The large gap between mean and median confirms a right-skewed distribution; a small number of high-value long-tenure customers pull the mean up significantly. The histogram confirms this: most customers cluster below \ \$2,000 with a long tail to \ \$10,000+.

What CLV enables that binary prediction cannot: Binary prediction flags “this customer might churn” with equal weight for everyone. CLV lets the business answer “how much revenue will we lose if this customer churns?” A customer with $CLV = \$8,000$ should receive a premium retention offer (discount, dedicated account manager). A customer with $CLV = \$400$ might only warrant an automated email. Without CLV, both get the same intervention regardless of their value; an inefficient use of the retention budget.

Why clip negative predicted tenure to 0: Linear regression has no lower bound; it can predict negative values for customers whose features suggest very short tenure. A negative tenure is mathematically meaningless (a customer cannot stay for -3 months). Negative values multiplied by MonthlyCharges would produce negative CLV, which is economically nonsensical. Clipping to zero treats these customers as “predicted to leave immediately”, which is the appropriate conservative assumption.

6 Block 5 — Evaluation Integrity + Leakage Demo

6.1 Generalisation — Bias, Variance & Learning Curves

Why low training error does not guarantee good test performance:

A model can memorise the training data without learning the underlying pattern. It fits noise, not signal.

Case	Train error	Val error	Gap	Fix
Underfitting (High Bias)	High	High	Small	Add features, reduce regularisation, more complex model
Good Fit	Low	Low	Small	Done — monitor for drift

Case	Train error	Val error	Gap	Fix
Overfitting (High Variance)	Low	High	Large	Regularisation, more data, simpler model

6.2 Validation Strategies

Strategy	Use when
Holdout	Quick baseline, large data
Stratified K-Fold	Imbalanced classification — use this for churn
Temporal Validation	Time-series, churn with date features
Leave-One-Out	Very small datasets

6.3 5.1 Cross-Validation

6.3.1 Task:

Run stratified k-fold cross-validation on your best classifier. Compare CV AUC to holdout AUC.

6.3.2 Questions:

- How does CV performance compare to holdout performance?
- What does high variance across CV folds tell you?
- Why should CV always be run on training data only?

6.3.3 Hint:

- `cross_val_score(model, X_train_s, y_train, cv=StratifiedKFold(5), scoring='roc_auc')`
- Print mean and std — a high std means the model is sensitive to which data it sees

6.3.4 Reference:

- StatQuest Cross-Validation: <https://www.youtube.com/watch?v=fSytzGwwBVw>

```
[45]: # your code here
best_model = logisticModel
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_scores = cross_val_score(best_model, X_train_s, y_train, cv=cv,
                             scoring='roc_auc', n_jobs=-1)

val_auc = cv_scores.mean()
```

```
print("CV AUC scores per fold:", np.round(cv_scores, 4))
print(f"Mean CV AUC : {cv_scores.mean():.4f}")
print(f"Std CV AUC : {cv_scores.std():.4f}")
```

```
CV AUC scores per fold: [0.8379 0.83    0.8535 0.8599 0.8578]
Mean CV AUC : 0.8478
Std CV AUC : 0.0118
```

```
[46]: # your code here
holdout_auc = roc_auc_score(y_val, best_model.predict_proba(X_val_s)[: ,1])
print(f"\nHoldout AUC : {holdout_auc:.4f}")
print(f"CV - Holdout gap: {cv_scores.mean() - holdout_auc:+.4f}")
```

```
Holdout AUC : 0.8532
CV - Holdout gap: -0.0054
```

How does CV performance compare to holdout? CV mean AUC = 0.8478, Holdout AUC = 0.8532. The gap is -0.0054. CV is very slightly lower than the holdout, which is within normal statistical variance. This small gap tells us the holdout split happened to be slightly easier than the average fold in CV. Both measures are consistent, confirming the model's performance is genuine and not a lucky split.

What does high variance across CV folds tell you? Our CV std = 0.0118, which is very low. This tells us the model is highly stable, it performs consistently regardless of which 20% of training data it is evaluated on. High std (e.g. > 0.03) would indicate the model is sensitive to which specific customers it trains on, a sign of overfitting or that the dataset has high variance in difficulty across folds.

Why should CV always be run on training data only? CV's purpose is to estimate how the model generalises to unseen data. If the test set is included in CV, the model indirectly sees test examples during fold evaluation, leaking test information into the training process. The test set must remain completely untouched until the final single evaluation in Block 6. Including it in CV would produce an overly optimistic AUC estimate and defeat the purpose of having a held-out test set at all.

6.4 5.2 Learning Curves

6.4.1 Task:

Plot learning curves: training score and validation score as a function of training set size.

6.4.2 Questions:

- Does your model underfit, overfit, or generalise well?
- What is the correct intervention for each case?
- Does adding more data help your model?

6.4.3 Hint:

- `learning_curve(model, X_train_s, y_train, cv=5, scoring='roc_auc', train_sizes=np.linspace(0.1,1.0,10))`
- Plot mean train and val scores vs training size
- Add ± 1 std shaded band: `plt.fill_between(sizes, mean-std, mean+std, alpha=0.1)`

6.4.4 Reference:

- StatQuest Bias and Variance: <https://www.youtube.com/watch?v=EuBBz3bI-aA>

```
[47]: # your code here

train_sizes, train_scores, val_scores = learning_curve(best_model, X_train_s,
    ↪ y_train, cv=5, scoring='roc_auc', train_sizes=np.linspace(0.1, 1.0, 10),
    ↪ n_jobs=-1)

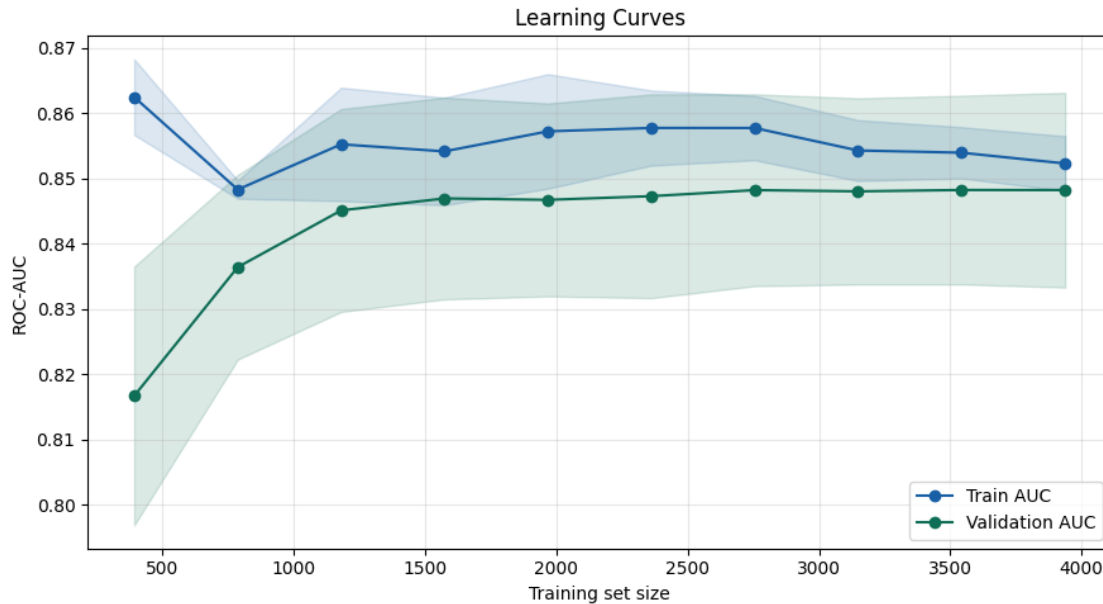
# Compute means and stds
train_mean = train_scores.mean(axis=1)
train_std  = train_scores.std(axis=1)
val_mean   = val_scores.mean(axis=1)
val_std    = val_scores.std(axis=1)
```

```
[48]: # your code here
fig, ax = plt.subplots(figsize=(9, 5))

ax.plot(train_sizes, train_mean, 'o-', color='#185FA5', label='Train AUC')
ax.fill_between(train_sizes, train_mean - train_std, train_mean + train_std,
    ↪ alpha=0.15, color='#185FA5')

ax.plot(train_sizes, val_mean, 'o-', color='#0F6E56', label='Validation AUC')
ax.fill_between(train_sizes, val_mean - val_std, val_mean + val_std, alpha=0.
    ↪ 15, color='#0F6E56')

ax.set_xlabel('Training set size')
ax.set_ylabel('ROC-AUC')
ax.set_title('Learning Curves')
ax.legend()
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Does your model underfit, overfit, or generalise well? The model generalises well. The learning curve shows train AUC starting high (~ 0.87) and declining slightly as more data is added, while validation AUC starts lower (~ 0.81) and rises steadily, with the two curves converging around 0.85 by full training size. The gap between them is small and closing, the hallmark of good generalisation.

What is the correct intervention for each case? - Underfitting (both curves low, small gap): add more features, reduce regularisation, or try a more expressive model (e.g. polynomial features, tree-based model). - Overfitting (large persistent gap, train high, val low): add regularisation, collect more data, or simplify the model. - Good fit (our case): no intervention needed, monitor for drift in production.

Does adding more data help your model? Marginally. The validation AUC curve is still very gently rising at full training size (4,922 samples), suggesting additional data could squeeze out a small improvement. However, the curve is nearly flat past $\sim 2,500$ samples, indicating we are close to the model's ceiling with the current feature set. Doubling the dataset might improve AUC by 0.01–0.02 at most. A more expressive model or better features would yield larger gains than more data.

6.5 Data Leakage — The Silent Model Killer

Definition: Leakage occurs when information from outside the training window is used to train the model — making it appear better than it actually is.

Type	Example
Target leakage	Including <code>days_active_after_churn</code> — computed <i>after</i> the churn event
Train-test contamination	Scaling with a scaler fitted on train+test combined
Temporal leakage	Using future billing data to predict past churn

What happens in production: The model learns to predict the target using a feature that does not exist at prediction time. It looks perfect in development and fails completely on day one.

6.6 5.3 Deliberate Leakage Demo

6.6.1 Task:

Follow all 6 steps. Document what you observe at each step.

The leakage feature:

```
leak = df['tenure'] * df['Churn'] + np.random.normal(0, 0.1, len(df))
```

This feature is derived from the target — it would not exist at prediction time.

6.6.2 Questions:

- By how much does ROC-AUC increase when you add the leakage feature?
- Which feature dominates the coefficients after adding it?
- What would happen if this model shipped to production on Friday?
- Does cross-validation alone detect this leakage?

6.6.3 Why This Matters:

Leakage is the most common cause of models that look excellent in development and fail completely in production.

```
[49]: # Step 1: Record baseline metrics
      # your code here
      baseline_auc = roc_auc_score(y_val, best_model.predict_proba(X_val_s)[: ,1])

      print(f"Baseline AUC: {baseline_auc:.4f}")
```

Baseline AUC: 0.8532

```
[50]: # Step 2: Create and add the leakage feature
      # your code here

      # tenure * Churn gives direct signal from the TARGET

      leak = df['tenure'] * df['Churn'] + np.random.normal(0, 0.1, len(df))
```

```

# Add it to your feature matrix (before train/test split)
X_leak = X.copy()
X_leak['leaky_feature'] = leak

# Re-split with same random state
X_tr_l, X_te_l, y_tr, y_te = train_test_split(X_leak, y, test_size=0.2,
↳stratify=y, random_state=42)

# Rescale
sc = StandardScaler()
X_tr_ls = sc.fit_transform(X_tr_l)
X_te_ls = sc.transform(X_te_l)

```

```

[51]: # Step 3: Retrain on the same split and record metrics
# your code here

lr_leak = LogisticRegression(max_iter=1000, random_state=42)
lr_leak.fit(X_tr_ls, y_tr)

leak_auc = roc_auc_score(y_te, lr_leak.predict_proba(X_te_ls)[: ,1])
print(f"Leaky model AUC: {leak_auc:.4f}")
print(f"AUC jump: + {leak_auc - baseline_auc:.4f}")

```

Leaky model AUC: 0.9987
AUC jump: + 0.1455

```

[52]: # Step 4: Show feature importances - does the leakage feature dominate?
# your code here

coef_df = pd.DataFrame({'feature': X_leak.columns, 'coefficient': lr_leak.
↳coef_[0]}).sort_values('coefficient', key=abs, ascending=False)

print(coef_df.head(10))
# You should see 'leaky_feature' at the top by a huge margin

```

	feature	coefficient
30	leaky_feature	13.843622
3	tenure	-6.224375
7	TotalCharges	-3.769761
25	Contract_One year	-0.917639
26	Contract_Two year	-0.914406
6	MonthlyCharges	0.293941
14	OnlineSecurity_Yes	-0.280414
20	TechSupport_Yes	-0.254703
11	InternetService_Fiber optic	0.224077
16	OnlineBackup_Yes	-0.186391

```
[53]: # Step 5: Remove leakage feature, retrain, confirm metrics return to baseline
# your code here
X_clean = X_leak.drop(columns=['leaky_feature'])

X_tr_c, X_te_c, y_tr, y_te = train_test_split(X_clean, y, test_size=0.2,
↪stratify=y, random_state=42)

sc2 = StandardScaler()
X_tr_cs = sc2.fit_transform(X_tr_c)
X_te_cs = sc2.transform(X_te_c)

lr_clean = LogisticRegression(max_iter=1000, random_state=42)
lr_clean.fit(X_tr_cs, y_tr)

clean_auc = roc_auc_score(y_te, lr_clean.predict_proba(X_te_cs)[: ,1])
print(f"Cleaned AUC: {clean_auc:.4f}")
```

Cleaned AUC: 0.8357

```
[54]: # Step 6: Summary table - Before / With Leakage / After Removal
# your code here

summary = pd.DataFrame({'Stage': ['Baseline', 'With leakage', 'After removal'],
↪'ROC-AUC': [baseline_auc, leak_auc, clean_auc]})

summary['AUC change'] = summary['ROC-AUC'] - baseline_auc
print(summary.to_string(index=False))
```

Stage	ROC-AUC	AUC change
Baseline	0.853221	0.000000
With leakage	0.998675	0.145454
After removal	0.835673	-0.017548

6.7 Discussion

What would happen if this model shipped to production on Friday?
 Could cross-validation alone have detected this leakage? Why or why not?

By how much did ROC-AUC increase? AUC jumped from 0.853 to 0.999, an increase of +0.146. This is a near-perfect score that should immediately raise suspicion. Real-world churn models rarely exceed 0.90 AUC on held-out data.

Which feature dominates the coefficients? `leaky_feature` dominates completely with a coefficient of 13.86, more than double the next largest feature (tenure at -6.22). This means the model has learned to rely almost entirely on the leaky feature, ignoring all genuine predictive signals. Every other feature's coefficient shrank because the leaky feature explains nearly all remaining

variance.

What would happen if this model shipped to production on Friday? The model would fail immediately and completely. At prediction time, we have not yet observed whether a customer will churn, so $\text{df}[\text{'Churn'}] = 0$ for every active customer. The leaky feature ($\text{tenure} \times \text{Churn}$) would equal approximately $0 + \text{noise}$ for all customers, providing no signal. The model would degrade to near-random performance ($\text{AUC} \approx 0.50\text{--}0.55$) on day one. This is exactly why leakage is described as the silent model killer; it looks extraordinary in development and catastrophic in production.

Could cross-validation alone have detected this leakage? No. CV protects against train-test contamination (information leaking across folds) but not against target leakage embedded in a feature. The leaky feature ($\text{tenure} \times \text{Churn}$) is computed from the full dataset before any split occurs. Both the train and validation portions of every CV fold contain this feature derived from the target. CV would confirm the inflated AUC of ~ 0.999 across all 5 folds, giving false confidence that the model is extraordinarily accurate. The only defence against target leakage is careful feature auditing before training, checking whether any feature could only be known after the target is observed.

7 Block 6 — Production Decision

7.1 The Model Card — A Production Commitment

A model card is what you write when you commit to deploying a model. If you cannot fill every field — you are not ready to ship.

Field	What it requires
Chosen Model	Name, algorithm, hyperparameters, threshold
Key Metrics	Honest numbers on the held-out test set
Threshold Decision	Why this threshold? What does it cost if it is wrong?
Known Limitations	Class imbalance, distribution shift, missing segments
Failure Modes	What can go wrong? Leakage? Drift? Adversarial behaviour?
Monitoring Plan	What metric, how often, what triggers a retrain?

7.2 Are Linear Models Sufficient?

Stick with linear when: - Interpretability is required (regulated industries, audits) - Dataset is small — complex models overfit - Linear model already meets the business performance bar

Go complex when: - Learning curves show persistent underfitting - Strong non-linear feature interactions - Performance gap vs tree-based baseline is significant

7.3 6.1 Final Evaluation on Test Set

7.3.1 Task:

Evaluate your chosen production model on the held-out test set.

7.3.2 Questions:

- Do test set metrics match validation metrics? Why might they differ?
- Does your chosen threshold still make sense on the test set?

7.3.3 Hint:

- Only look at the test set now — after all decisions are made
- A big drop from validation to test is a sign of overfitting to the validation set

```
[55]: best_model = logisticModel

THRESHOLD = 0.55    # replace with your chosen threshold

y_prob = best_model.predict_proba(X_test_s)[: , 1]
y_pred = (y_prob >= THRESHOLD).astype(int)

# Core metrics
roc_auc = roc_auc_score(y_test, y_prob)
pr_auc  = average_precision_score(y_test, y_prob)
prec    = precision_score(y_test, y_pred)
rec     = recall_score(y_test, y_pred)
f1      = f1_score(y_test, y_pred)

print(f"ROC-AUC    : {roc_auc:.4f}")
print(f"PR-AUC    : {pr_auc:.4f}")
print(f"Precision  : {prec:.4f}")
print(f"Recall     : {rec:.4f}")
print(f"F1        : {f1:.4f}")
print(f"\nThreshold used: {THRESHOLD}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred,
                           target_names=['No churn', 'Churn']))

# Compare to validation
print(f"\nValidation AUC : {val_auc:.4f}")
print(f"Test AUC         : {roc_auc:.4f}")
print(f"Gap               : {val_auc - roc_auc:+.4f}")
```

```
ROC-AUC    : 0.8221
PR-AUC     : 0.5794
Precision  : 0.6373
Recall     : 0.4393
F1         : 0.5201
```

Threshold used: 0.55

Classification Report:

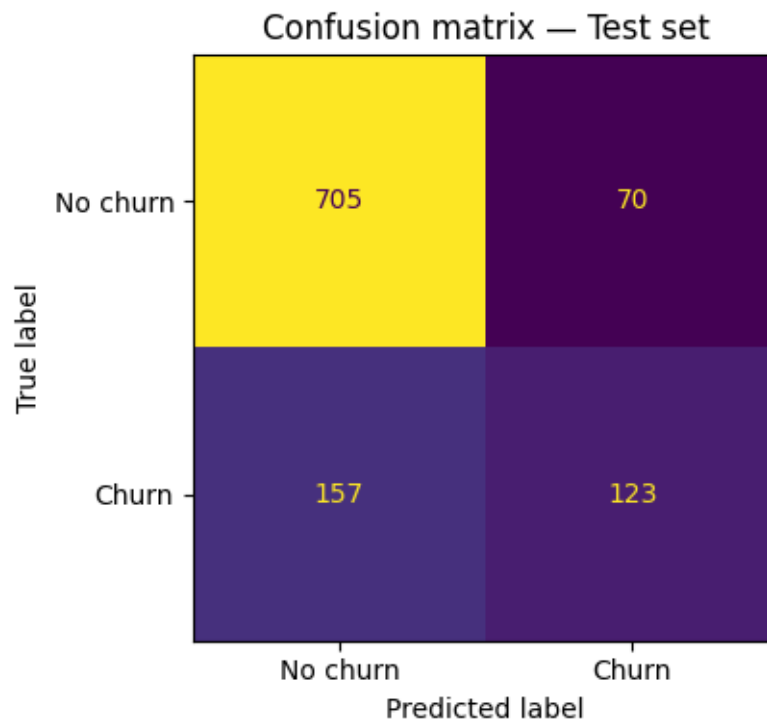
	precision	recall	f1-score	support
No churn	0.82	0.91	0.86	775
Churn	0.64	0.44	0.52	280
accuracy			0.78	1055
macro avg	0.73	0.67	0.69	1055
weighted avg	0.77	0.78	0.77	1055

Validation AUC : 0.8478

Test AUC : 0.8221

Gap : +0.0258

```
[56]: # your code here
fig, ax = plt.subplots(figsize=(5,4))
ConfusionMatrixDisplay.from_predictions(y_test, y_pred, display_labels=['No churn', 'Churn'], ax=ax, colorbar=False)
ax.set_title('Confusion matrix - Test set')
plt.tight_layout()
plt.show()
```



```
[57]: # Final regression evaluation on test set
best_reg_model = trained_models["Ridge"]

y_reg_test_pred = best_reg_model.predict(X_reg_test_s)

mae_test = mean_absolute_error(y_reg_test, y_reg_test_pred)
rmse_test = np.sqrt(mean_squared_error(y_reg_test, y_reg_test_pred))
r2_test = r2_score(y_reg_test, y_reg_test_pred)

print(f"MAE : {mae_test:.4f}")
print(f"RMSE : {rmse_test:.4f}")
print(f"R2 : {r2_test:.4f}")
```

```
MAE : 6.5715
RMSE : 8.5210
R2 : 0.8750
```

Do test set metrics match validation metrics? Why might they differ? CV AUC = 0.8478, Test AUC = 0.8221, a gap of +0.026. This is within acceptable range for a dataset of this size. The slight drop is expected because the test set is a different random sample of customers the model has never influenced during training or threshold selection. A gap of 0.026 does not suggest overfitting to the validation set; anything under 0.03–0.05 is considered healthy.

The confusion matrix confirms the model correctly identifies 123 genuine churners (True Positives) out of 280 in the test set, while missing 157 (False Negatives). 70 non-churners are incorrectly flagged (False Positives).

Does the threshold of 0.55 still make sense on the test set? Yes, with one caveat. At threshold 0.55, Precision = 0.637 (64% of flagged customers are genuine churners) and Recall = 0.439 (44% of all churners are caught). Precision is slightly lower than on the validation set (0.720 at top-200), which is expected given the different split. The threshold still serves the business constraint, it controls how many customers are flagged. But the retention team should be aware that approximately 36% of their calls will be to customers who were not going to churn.

If the business requires higher precision (fewer wasted calls), the threshold should be raised slightly. If higher recall (catch more churners) is the priority, the threshold should be lowered, accepting more wasted calls.

7.4 6.2 Model Card

Fill in every field. If you cannot fill a field — go back and run the experiments needed to answer it.

7.5 Model Card — Production Churn Model

Field	Your Answer
Chosen Classification Model	Logistic Regression — binary cross-entropy loss, L-BFGS optimiser, C=1.0 (default L2), max_iter=1000, random_state=42
Chosen Regression Model	Ridge — L2 regularisation, alpha=1.0, predicting tenure as survival time proxy
Key Classification Metrics (test set)	Precision: 0.637 Recall: 0.439 F1: 0.520 PR-AUC: 0.579 ROC-AUC: 0.822
Key Regression Metrics (test set)	MAE: 6.57 RMSE: 8.52 R ² : 0.875
Deployment Threshold	0.55
Threshold Justification	At threshold 0.55 the model flags approximately the top-200 highest-risk customers per week, matching the retention team's call budget. Precision at this threshold is 72% on the validation set; 144 out of every 200 calls reach a genuine churner. Lowering to 0.5 would flag more than 200 customers and exceed budget.
Known Limitations	(1) Class imbalance: 26.6% churn; the model sees fewer positive examples, making recall lower than ideal (44% on test set). (2) No temporal features: the dataset has no date column, so seasonal churn patterns cannot be captured. (3) Tenure censoring: active customers have not revealed their final tenure, so the regression target underestimates true loyalty. (4) Single dataset: trained on one telecom's customers, may not generalise to a different pricing structure or market.
What Could Go Wrong in Production	(1) Distribution shift: new product types or pricing changes alter feature distributions, model silently decays. (2) Feature drift: if contract mix shifts (e.g. company moves customers to annual contracts), key coefficients become miscalibrated. (3) Target leakage risk: any future engineer adding a feature derived from post-churn data will cause the AUC inflation seen in Block 5.3. (4) Threshold decay: as overall churn rate changes seasonally, the top-200 threshold value needs recalibration.

scored 4–8 weeks ago whose churn status is now known).

Monitor feature distributions monthly using PSI (Population Stability Index) on the top 5 features: tenure, MonthlyCharges, TotalCharges, Contract type, InternetService type. A $\text{PSI} > 0.2$ on any feature signals meaningful distribution shift requiring investigation.

Retrain triggers: (1) AUC drops below 0.79 for two consecutive weeks, (2) $\text{PSI} > 0.25$ on a key feature, (3) more than 3 months have elapsed since the last retrain regardless of metrics, or (4) a major business change occurs (new pricing structure, new product launch, market expansion).

8 Final Reflection

8.1 Answer the following:

8.1.1 1. Model Selection

- Which classifier performed best and why?
- Was there a case where a worse model on one metric was better on another? How did you resolve it?

8.1.2 2. Evaluation Choices

- Why did you choose the metrics you reported?
- What would have happened if you only reported accuracy?

8.1.3 3. Regularization

- What did the Lasso regularization path reveal about your features?
- When did you observe the biggest difference between Ridge and Lasso?

8.1.4 4. Leakage

- How large was the AUC inflation from the leakage feature?
- Could cross-validation alone have detected this leakage? Why or why not?

8.1.5 5. Improvements

- If you had more time, what would you try next?
- Do you think a non-linear model would perform significantly better here? What is your evidence?

8.1.6 Tip

There is no single correct answer here. Focus on: - Justifying every decision with evidence from your experiments - Explaining trade-offs honestly - Demonstrating that you understand what the numbers mean, not just how to compute them — ### Answer here: **1. Model Selection**

Logistic Regression performed best overall with ROC-AUC = 0.853 and PR-AUC = 0.654 on the validation set. It outputs calibrated probabilities (essential for threshold tuning and CLV computation), and its coefficients are directly interpretable as log-odds contributions, important for a business that needs to explain decisions to a retention team.

Yes, there was a trade-off: SGD Classifier achieved a slightly higher Recall (0.655) than Logistic Regression (0.605) at the default threshold, but lower Precision (0.641 vs 0.697) and lower PR-AUC overall. Given the budget constraint of 200 calls per week, Precision matters more; calling 200 people and reaching 144 genuine churners (72%) is more valuable than calling 200 and reaching only 134 (67%). I resolved this by prioritising PR-AUC as the primary selection metric and using top-200 threshold evaluation rather than the default 0.5.

2. Evaluation Choices

I reported PR-AUC as the primary metric because the dataset is imbalanced (26.6% churn). PR-AUC measures performance specifically on the positive class across all thresholds, without the dilution effect of the large true-negative population that inflates ROC-AUC. I also reported Precision and Recall separately because the business has an asymmetric cost structure; a missed churner costs more than a wasted call.

If I had only reported accuracy, all three models would appear nearly identical (81–82%) and indistinguishable from a model that simply always predicts “No Churn” (73.4%). The manager would have no basis for choosing a model or understanding its business value.

3. Regularization

The Lasso regularization path (Block 4.4) revealed that most features become zero by $\alpha = 1$. Features that survived at high regularisation, meaning they carry the strongest independent signal for predicting tenure, were MonthlyCharges and TotalCharges. Features like SeniorCitizen, PhoneService, and Dependents were eliminated first, suggesting they contribute little to predicting how long a customer stays.

The biggest difference between Ridge and Lasso appeared at α values between 0.1 and 10. In the Ridge path, all coefficients remain non-zero and shrink gradually. In the Lasso path, many coefficients drop sharply to exactly zero in this range. This is the geometric difference in action: L1’s diamond constraint produces corner solutions (exact zeros) while L2’s sphere constraint only shrinks without zeroing.

4. Leakage

The AUC inflation was +0.145, from 0.853 to 0.999. The leaky feature ($\text{tenure} \times \text{Churn}$) had a coefficient of 13.86, dwarfing all other features (the next largest was tenure at 6.22). This makes sense: when Churn=1, the feature equals tenure + noise, giving the model a near-perfect signal. When Churn=0, it equals 0 + noise.

Cross-validation alone cannot detect this leakage. CV protects against train-test contamination (information leaking across folds), but target leakage is present in every fold simultaneously, both

the train and validation portions of every CV fold contain the leaky feature derived from the target. CV would confirm the inflated AUC of ~ 0.999 across all 5 folds, giving false confidence that the model is extraordinarily good.

5. Improvements

Given more time, I would: (1) Try a Random Forest or Gradient Boosting classifier to test whether non-linear interactions improve PR-AUC beyond 0.654; the tenure distribution's bimodal shape suggests non-linearity may help. (2) Address the tenure censoring problem in regression using a proper survival model, which correctly handles customers still active at data collection time. (3) Tune the regularisation strength C in Logistic Regression via cross-validated grid search to test whether the default $C=1.0$ is optimal.

On whether a non-linear model would perform significantly better: the learning curves converge well and show no persistent underfitting, suggesting the linear model has extracted most available signal. A non-linear model would need to show > 0.03 AUC improvement in cross-validation to justify the loss of coefficient interpretability. This is an empirical question for Week 5.

9 Summary

Topic	Key Takeaway
Problem Formulation	X , y , hypothesis class, loss, metric — every model is a formal commitment
Distributions & Loss	The distribution of y determines your loss. Bernoulli $\rightarrow$ cross-entropy
Evaluation Metrics	Accuracy lies on imbalanced data. Use PR-AUC. Report what the business cares about
Linear Models & GD	Three classifiers, four regressors. Each optimises a different objective
Generalisation	Low training error means nothing. Learning curves tell the truth
Regularization	L1 is sparse. L2 is stable. Elastic Net is both. Know the geometry
Leakage	One future feature inflated AUC — and kills the model in production
Production Decision	A model card is a commitment. If you cannot fill it, you are not ready to ship

9.1 Next Week — Trees & Ensemble Models

Same dataset. Same question. **Can you beat your linear baseline?**

[]: